

What kind of thing is a planet?

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Contents

1	Introduction	1
2	Review and Theory	2
2.1	Planets from first principles	2
2.2	Heat and geometry: planetary basics	3
2.3	From planetary to exoplanetary: an expanding scope of inquiry.	5
2.3.1	Local planets	5
2.3.2	Exoplanets	7
2.4	Geodynamics: an evolving discipline	8
2.4.1	Empirical foundations of geodynamic thought	8
2.4.2	Challenges to the conventional model of plate tectonics	9
2.4.3	Continents: a decisive control in the tectonic system	11
2.4.4	Tectonic modes: beyond plate tectonics	12
2.4.5	Tectonic mode transitions and planetary memory	14
2.5	Review	16
3	Foundations	17
3.1	The analytical toolkit	17
3.1.1	The Nusselt number	17
3.1.2	The Prandtl, Grashof, Reynolds, and Rayleigh numbers	18
3.1.3	Linear stability analysis and the critical <i>Rayleigh</i> number	20
3.1.4	Boundary layer theory and the $Ra - Nu$ scaling	22
3.1.5	Chaos and attraction: approximate solutions to insoluble equations	27
3.2	Incorporating radiogenic heat: the internal- and mixed-heating cases	29
3.2.1	Internal heating	30
3.2.2	Mixed heating	32
3.3	Thinking outside the box: building a cylindrical domain	34
3.3.1	Establishing a coordinate system	34
3.3.2	Conduction in the basally-heated cylindrical case	37
3.3.3	Internal heating in the annulus	41
3.3.4	Mixed heating in the annulus	42
3.4	Numerical methods and the Underworld code.	44
3.4.1	Numerical methods	45
3.4.2	The Underworld code	46
3.4.3	Underworld in the annulus	48
3.5	Conclusion	48
4	Methods	49
4.1	The crisis of dimensionality	50
4.1.1	Mass	50
4.1.2	Space	51
4.1.3	Time	51

5	Critical	53
5.1	Conclusion	53
6	Temperature-dependant rheology	54
7	Viscoplastic rheology	55
8	New techniques	56
9	Discussion	57
10	Conclusion	58

Chapter 1

Introduction

An introduction.

Chapter 2

Review and Theory

The purpose of this study is to deduce what underlying parameters most strongly control a planet's choice of tectonic mode. The fundamental question regards the existence or otherwise of a well-defined 'tectonic phase diagram' in which primarily temperature defines the proper global geodynamic regime at any given time. Phase spaces are defined by their phase transitions, and in the field of planetary geodynamics, only one such transition is well-attested and amenable to study: the emergence on Earth of plate tectonics from an earlier pre-tectonic state of disputed character. The Early Earth is therefore key to the larger question of tectonic modality.

In this review chapter, we will present a brief history of scholarship on the topic, summarise the chief lines of evidence, and discuss the leading hypotheses regarding the divergent fates of Earth and our most familiar planetary cousins. Our line of inquiry will be illustrated with some simple one-dimensional modelling. We will not discuss in detail the analytical and numerical intricacies of the problem at hand, which will instead be related as appropriate in the prefacing remarks of the relevant results chapters (Chapters 3, 4, 5, & 6); nor will we provide much background or context for our methodological approach, which is treated separately (Chapter 2 & Chapter 7).

2.1 Planets from first principles

Our subject is planets - but what is a planet?

According to the International Astronomical Union highly-publicised ruling of 2006 [[International Astronomical Union, 2006](#)], "a planet is a celestial body that; 1) is in orbit around a sun, 2) has sufficient mass to have a nearly round shape, 3) has "cleared the neighborhood" around its orbit." Under this rubric, Pluto, Charon, Eris, and other icy worlds of the outermost solar system are disqualified. Indeed, the rationale for the IAU's opinion was that any broader definition would result in a categorical calamity: potentially hundreds of new worlds would achieve the mantle of 'planet', including many that at the time seemed unlikely to be of much significance or interest [[Sarma et al., 2008](#)], creating needless public and academic 'annoyance' [[Basri and Brown, 2006](#)].

The IAU's ruling sparked an impressive global backlash, including an attempt by lawmakers to quash the revision [[Zielinski, 2007](#)] and various pleas for celebrity astronomers to intercede [[Tyson, 2009](#)]. The case has even become a niche topic of sociological literature, with papers discussing its effect on children [[Jarman and McClune, 2009](#), [Broughton et al., 2013](#)] and its impact on popular perceptions of the credibility of astronomy [[Christensen, 2007](#), [Messeri, 2010](#)].

In the immediate aftermath, a petition of over 300 signatures from attendees at the IAU meeting itself protested the new classification, which had been decided by a partial convention of the union rather than a general colloquium, perhaps to avoid protest [[Cartlidge, 2006](#)]. Alan Stern, the director of the New Horizons mission to the outer solar system, quickly became one of the highest-profile advocates for revising or ignoring the IAU opinion [[Hogan, 2006](#)]. He and his colleagues have since developed a so-called 'geophysical definition' which is now the standard at NASA [[Runyon et al., 2017](#)]: "A planet is a sub-stellar mass body that has never undergone nuclear fusion and that has sufficient self-gravitation to assume a spheroidal shape adequately described by a triaxial ellipsoid regardless of its orbital parameters."

Stern's definition is a profound revision of the cosmos, but one which is well-argued and well-suited to the purposes of planetary scientists. Under Stern's mandate, not only Pluto and Eris, but also the asteroid belt objects Ceres and Vesta, as well as various 'satellite planets' - Earth's moon, Pluto's partner Charon, and most of the satellites of the outer gas giants - all would be considered equally as planets. Such an expansion puts the geophysical, atmospheric, and climatic aspects of planets front and centre, above and beyond their orbital or formational histories. It sets the stage epistemologically for a fundamental deepening of our understanding of the unique processes that govern sub-stellar objects in our universe, and - as we will discuss later (Chapter 8) - may be regarded in retrospect as a milestone event in the birth of a new field of focussed, integrated, holistic planetary science.

2.2 Heat and geometry: planetary basics

The geophysical definition of a planet provides some clarification on what should and what shouldn't be included as essential considerations of our topic. A planet is a gravitationally-bound ellipsoidal object, not currently or formerly made of plasma, surrounded by mostly empty space. Such a definition has some immediate implications. The chemical and gravitational prescriptions of Stern's definition provide an upper and lower spatial scale respectively: anywhere from the radius of Mimas, at 200 km the smallest rounded body in the solar system [Lodders and Fegley, 1998], to the radius of Jupiter, about 70,000 km, above which any additional mass tends to be compressed without increasing planet size [Basri and Brown, 2006]. It implies a temporal scale too: between the atmospheric (hours) and geological (gigayear) timescales.

Other implicit constraints are more subtle. Because a planet is qualified by its shape - a function of mass and matter - it is parsimonious to stipulate that these quantities are connate to each planet and cannot be changed. Of course, in reality, such changes do occur: consider the formation of Earth's moon from a putative impact with a Mars-sized planetoid in the Hadean [Canup, 2012], or the hypothesised delivery to Earth of a significant mass of volatiles by icy planetoids [Owen and Bar-Nun, 2000]. In the system outlined here, however, such events would be considered 'catastrophic' - outside of the prescribed spatiotemporal scales of the subject, and violating the prescribed conservative quantities of the problem; in essence, such a calamity would be considered the 'death' of the original planet and the 'birth' of a new one.

There are also more sustained violations: erosion of planetary atmospheres into space, for example [Perez de Tejada, 1992, Kulikov et al., 2006], or the capture of solar protons in the aurorae [Rees, 1982], or the ongoing low-level exchange of meteoroids between Earth and Mars [Melosh and Tonks, 1993]. Such endemic exceptions, however, plague all systems; hence we will dub them 'pathological', requiring careful handling but not deference.

Planets, then, can be said to be spheroidal, sub-stellar objects of a fixed mass and bulk chemistry. Within these parameters, however, there are considerable degrees of freedom:

- The geometric characters of the spheroid can change - eccentricity, radius, and obliquity - with consequences for the centre of mass, the gravitational field, and the tidal coupling of the planet with any extra-planetary systems.
- The distribution of mass can change, anywhere between the extremes of an arbitrarily thin hollow shell and an arbitrarily dense core, with consequences for the moment of inertia and rate of rotation as well as the gravitational field and the gravitational binding energy. The total mass, of course, cannot change - this would be 'catastrophic' - hence the gravitational pull of a planet at a sufficient distance above the surface is fixed.
- The angular velocity of the planet can change due to internal forces and geometric transformations. Angular momentum can also change, though necessarily only by interactions with extra-planetary bodies: when these interactions are periodic, systematic, and persistent over geological time, they may be parameterised as an intrinsic forcing within the planet; otherwise, they may be considered 'pathological' (if frequent) or 'catastrophic' (if infrequent) exceptions, outside the intrinsic scope of the problem.
- The chemical makeup of the planet can change. Indeed, all the degrees of freedom within any constituent molecules of a planet are available as degrees of freedom of the planet as a whole, as

are all the possibilities for the breaking of old bonds and the forging of new ones.

- The isotopic makeup of a planet can change, albeit largely without consequence for bulk chemistry - hence their enormous analytical utility as tracers and monitors for Earth sciences. Earth is overwhelmingly composed of less than 300 stable nuclides [Thoennessen and Sherrill, 2011].
- The elemental inventory of a planet can change; here, however, we must be careful. The decay of short-lived radioactive nuclides like Uranium-235 and Thorium-232 into daughter elements are significant to the system; the decays of extremely long-lived nuclides like Uranium-238 and Calcium-48 are not significant over ‘planetary’ timescales as stipulated above (“Nudat 2” n.d.). Decay of non-radioactive nuclides due to the finite half-lives of subatomic particles are not significant on any reasonable timescale and are not available as degrees of freedom for planetary processes as we choose to consider them.
- The spatial distribution of chemical and elemental species can change: to wit, the differentiation of the felsic crust, mafic mantle, and metallic core due to gravitational forcing; but also, the partitioning of rare-earth elements and radiogenics into the continental crust due to purely chemical factors.

Each of these degrees of freedom entail certain endothermic or exothermic reactions, and these naturally chain one to another in complex ways. The net result, however, is the exhaustion of all possible gradients and the conversion of all potential energy into diffuse kinetic energy, i.e. heat. But what is the total internal energy of a planet - its enthalpy?

Because there is no such thing as ‘absolute’ enthalpy, it is necessary to first define a natural point of reference for the system against which changes in the energy budget may be compared. Thankfully, the stipulations we have already made regarding the conserved quantities of planets, as well as the limitations we have provided on relevant spatiotemporal scales and the resultant pruning of available degrees of freedom, imply one fairly reliable, though somewhat counter-intuitive, reference point: not the planet’s initial condition, but rather its ‘rest state’ after a sufficient span of time.

Consider a situation where the Earth as a whole exchanges no heat with its environment. In fact, this is a surprisingly credible simplification: the global geothermal flux is estimated at only 47 terawatts [Lay et al., 2008, Davies and Davies, 2010], predominantly along the mid-ocean ridge [Davies, 2013]. This is insignificant compared to the 100,000 terawatts processed by the solar/atmospheric/oceanic system and comparable to the over 16 terawatts consumed by human civilisation in the modern era [IEA, 2007]; a small-enough flux that, if held constant and ignoring all internal heat production, a billion years would only see a 232 K drop in planetary average temperatures [Sleep, 2000]. Indeed, looking backward, we find Earth’s core has only lost in the order of 50 K per billion years since the planet was formed [Verhoogen, 1980], confirming our intuition.

What is the ‘ultimate’ temperature of an isenthalpic planet such as we have described? This will be the temperature obtained at an arbitrarily remote time in the future - long enough for all permissible degrees of freedom to be exploited. At this time, only heat remains: a planet has no more tricks to perform. It transpires that this number is not all that difficult to derive, nor so uncertain as we might assume - at least in comparison to the inverse problem, which is fraught with controversy.

On ‘Ultimate Earth’, the radiogenic nuclides have completely decayed into heat and stable daughters, while the stable nuclides have sorted by density from top to bottom - a crude but serviceable assumption. The Earth is now a perfect sphere of the smallest possible size, and its rotation has completely ceased due to tidal interactions with other cosmic masses.

We attempted to calculate the temperature of Ultimate Earth based off order-of-magnitude estimates, published scaling laws, and reference data [Dziewonski and Anderson, 1981, Lodders and Fegley, 1998, Sleep, 2000, Korenaga, 2003, De Pater and Lissauer, 2015, Zeng et al., 2016]. When calculated from present day observations, we derive a temperature of between 4,000 and 5,000 K, or about 2.92×10^{31} joules: comparable to the temperature of the Earth’s core, where most of today’s heat is stored [Jaupart and Mareschal, 2010]. When we instead venture to calculate the temperature starting from the presumed initial condition of the Earth, we obtain an Ultimate Earth temperature an order of magnitude higher - 40,000 to 50,000 K, about 2.75×10^{32} joules. In other words, the Earth’s heat has fallen by a factor of ten over its 4.5 billion year history.

The reason we can be fairly confident about these figures is that they are dominated by gravitational

potential energy and the heat of accretion, which can both be trivially derived analytically from Earth's mass; cumulatively, these factors account for about 10^{32} joules of Ultimate Earth's heat budget, about 93%. Some 88% of this energy is converted instantly into heat upon accretion; another 9% is released through differentiation, of which 8.7% is due to the core and 0.27% is due to the crust. Only 3% of the Earth's original gravitational binding energy remains as potential energy which could feasibly be released as work and heat. The remaining degrees of freedom - orbital, chemical, and nuclear - add up to a little over 10^{31} joules, but together make up the majority of the remaining energy available to be released. Interestingly, the amount of heat that could theoretically be liberated by gravitational sorting of nuclides - i.e. the formation of continents - is comparable to the remaining radiogenic budget of the Earth. If as much continental volume were to be produced in the next 5 Gy as in the previous, it would represent a significant proportion of geothermal flux.

In reality, the Earth is not isenthalpic: it receives solar heat and sheds geothermal heat into space. We observe here that some 89% of the Earth's heat has been lost to space since formation. This number is very close to the 88% of heat that would have been released geologically instantaneously at accretion. As Lord Kelvin famously calculated [Thomson, 1862], the maximum timescale to reach a present-day geotherm from a wholly molten state is a mere tens of millions of years, and much, much shorter than that if more active thermal processing is at work. Hence it is conceivable, though unverifiable, that almost all of Earth's accretionary heat was lost shortly after formation, with freezing of the core and radiogenic decay since accounting for most if not all of the observed geothermal fluxes at magnitudes similar to today's.

We have laid out what is admittedly a fairly rudimentary construct to commence our thesis. However, we argue it furnishes the appropriate 'ballpark' figures and situates the intuition in the proper place for a truly planetary way of thinking about thermodynamics and flow. The essential lesson of the Ultimate Earth concept is that the overall thermal regime of a planet is in no hurry to reach equilibrium with space; that in fact, over the lifetime of many planets - curtailed by the circumstances of their host stars to mere billions, rather than tens of billions of years - no state resembling Ultimate Earth will ever be achieved by star-bound worlds. In short, heat is not destiny for a planet.

2.3 From planetary to exoplanetary: an expanding scope of inquiry

It is incredible to consider that, within the lifetime of a single PhD student, a family of nine known planets has become a universe of thousands.

The expansion in scope of planetary sciences has occurred so rapidly it is easy to lose track of what has changed. Discoveries, in our solar system and beyond, are moving faster than theory, and it seems a major revolution in our understanding of planets waits to be realised. An atmosphere of genuine hope and excitement surrounds the field - both within the academy and beyond.

Here we present a brief overview of relevant recent discoveries in our home solar system and beyond.

2.3.1 Local planets

Unmanned exploration of our local system is entering a renewed era of great activity, emboldened by growing public interest and increasing private cash [Baker and Chandran, 2018]. This exploration is yielding provocative evidence that challenges our understanding of planetary dynamics.

While much of note has been uncovered on the rocky planets of our inner solar system, especially Mars, Venus, and the Moon, here we will discuss more recent discoveries whose import has arguably not yet been fully recognised in the broader geodynamics literature - planetary bodies that were supposed to be inert, which have proved to be full of activity. On the moons of Jupiter and Saturn, and far out beyond the orbit of Neptune, young, dynamic surfaces are being revealed, speaking to active internal processes that may support life - and hold invaluable clues to untangling the history of Earth and the deeper questions of tectonics.

Europa Jupiter's moon Europa has proved to be surprisingly analogous to Earth: its surface ice has been shown to exhibit plate-like tectonics [Patterson et al., 2006], with familiar fault patterns [Ay-

din, 2006], deformation at fault tips [Kattenhorn and Marshall, 2006], cryovolcanic ‘igneous’ terrains [O’Brien et al., 2002], strikingly Earth-like ridge flexures, forebulges, and other collisional features [Hurford et al., 2005], and some very recent evidence of ongoing subduction [Arculus et al., 2019].

Europa is heated internally by tides [Hussmann and Spohn, 2002], but beyond that, little is known of its thermal state. Some have speculated the existence of deep sea vents which could be excellent candidates for hosting life [Lipps and Rieboldt, 2005]. Whether Europa proves fertile or not, it remains the only known example of active horizontal tectonics beyond Earth, and hence is of tremendous importance to the broader question of what makes a ‘living’ planet.

Io If Europa provides some context for the Earth of today, another of Jupiter’s moons, Io, may hold the key to its past. Io, which is comparable in size and makeup to our Moon [Kuskov and Kronrod, 2000], is the only known body beyond Earth proven to exhibit non-icy volcanism in the present day [Moore et al., 2007]. The scale of volcanism on Io is extreme and unmatched in the solar system [McEwen et al., 2000]; first observed during the Voyager [Carr et al., 1979] and Galileo [McEwen et al., 1998] missions, the New Horizons flyby in 2007 detected no fewer than eleven active ejecta plumes, some jetting higher than 350 km [Spencer et al., 2007]. Io’s tremendous activity is related to its great internal heat, derived from intense tidal forces [Bart et al., 2004], which may even be sufficient to generate a subsurface magma ocean [Tyler et al., 2015].

Though initially suspected to be relatively low-temperature due to sulphur content [Smith et al., 1979], Io’s lavas are now thought to be mostly ultramafic and extremely hot [McEwen et al., 1998, Davies et al., 2008], fuelling analogies with the Archaean komatiite lavas on Earth [Williams et al., 2000, Nna Mvondo and Martinez-Frias, 2007]. For these reasons, Io - and the supposed ‘heat pipe’ tectonic mode that operates there (discussed later) - may provide a critical window into the early Earth [Matson et al., 1998, Moore and Webb, 2013, Kankanamge and Moore, 2016, Moore et al., 2017].

Enceladus Saturn’s moon Enceladus may be second only to Earth in geological activity [Spencer et al., 2009, Czechowski, 2017], making it an indispensable reference for comparative planetology.

The dynamic nature of Enceladus first became known with the Voyager missions, when its surface was found have large, crater-free terrains [Hanel et al., 1982, Smith et al., 1982] interpreted as evidence of cryovolcanism [Kargel and Pozio, 1996], as well as some apparently tectonic features such as grabens and fault banding [Squyres et al., 1982], arguing for the existence of ongoing, active resurfacing on this small, frigid moon [Squyres et al., 1983] where none had been considered possible before [McKinnon, 1987]. Enceladus was a chief target of the later Cassini mission, which captured high-resolution photos of the moon’s young and fault-marked surface [Roatsch et al., 2008], acquired magnetometric evidence of a volcanically-replenished water-derived atmosphere [Dougherty et al., 2006], and, most impressively, observed a large cryovolcanic plume [Ostro et al., 2006] composed of almost pure water [Waite et al., 2006] jetting from a tectonically active region around the south pole [Porco et al., 2006].

Enceladus is an ‘ice moon’, with a silicate core overlain by a mostly watery mantle and a water ice outer crust [Iess et al., 2014]. Its tiny size - less than 500 km in diameter - makes its evident activity a conundrum [Spencer et al., 2009]. Like Europa and Io, Enceladus’ internal heat had been presumed to be predominantly tidal [Nimmo et al., 2014], dissipated almost exclusively throughout the icy mantle [Roberts and Nimmo, 2008]; however, measured heat fluxes and dynamic modelling demand alternative sources, thought to be supplied by significant radiogenic heating from the silicate core [Schubert et al., 2007]. Consequently, Enceladus - like Earth - has both volumetric and basal heating components, and hence a Urey number, though no published estimates for this quantity yet exist. Also, on Enceladus, like on Earth, a feedback exists between the two heat sources: while Earth core heat plays a role in redistributing and partitioning heat-producing elements in the silicate mantle, Enceladus’ radiogenic heat may shift and enhance the tidal dissipation forces in its icy mantle - which may explain why this particular moon is full of activity, while the icier but otherwise similar Mimas is inert [Schubert et al., 2007].

Enceladus’ tectonics have been variously characterised as Earth-like plate tectonics [Kargel, 1983], Venusian episodic overturn [Tobie et al., 2008, O’Neill and Nimmo, 2010], and mobile-lid convec-

tion [Barr, 2008]. Particular analogues aside, what Enceladus argues above all is that the laws of geodynamics are not intrinsic to any particular chemistry or any particular spatiotemporal scale.

Pluto Pluto, controversially relegated to ‘dwarf’ status by the International Astronomical Union [Marschall and Maran, 2009], has since become the latest overlooked body to exhibit surprising geological activity [Hand, 2015]. Far from being a glorified comet, the New Horizons flyby in 2015 [Stern et al., 2015] showed Pluto to be a living world [Moore et al., 2016], with dunes [Telfer et al., 2018], troughs [Grundy et al., 2016], six kilometre-tall mountains and plateaux [Schenk et al., 2018], active glaciology [Bertrand and Forget, 2016], cryovolcanoes [Singer et al., 2018] and volcanic resurfacing [Singer et al., 2017, Cruikshank et al., 2019, Dalle Ore et al., 2019], and a thin but influential atmosphere [Gladstone et al., 2016]: all in all, a range of features that collectively “rival Mars in richness” according to one research team [Stern et al., 2018].

Most significantly, it now seems Pluto - like the other icy bodies here discussed - may possess a subsurface water ocean of its own [Nimmo et al., 2016]; seasonal partial freezing of that ocean might explain the extensional features on the surface [Hammond et al., 2016] and drive some of the observed cryovolcanism [Dalle Ore et al., 2019]. However - unlike Europa, Io, or Enceladus - tidal heat cannot be a factor on Pluto. This has led to speculation that the icy moon is anomalously well-insulated by clathrates [Kamata et al., 2019]. In any case, the vitality of Pluto’s active tectonics despite the moon’s small size and presumably meagre thermal inventory underscores that tectonics can achieve a lot with a little over geological timescales.

2.3.2 Exoplanets

Since the first confirmed sightings of planets beyond our solar system in the 1990s [Wolszczan and Frail, 1992, Mayor and Queloz, 1995] the number of known exoplanets has climbed well over 4,000 [NASA Exoplanet Science Institute, 2019]. Most of this extraordinary success is owed to the Kepler mission, now terminated [Adams et al., 2012]. Some of these worlds have local analogues; many do not [De Pater and Lissauer, 2015]. Planets have been observed around stars of all sizes, ages, and temperatures [Burrows and Marcy, 2014], and hundreds of multiple-planet systems have been identified, including some with known populations comparable to our own solar system [Tóth and Nagy, 2014, Gillon et al., 2017]. Statistical analyses which cross-correlate hit-rate with observation probability have concluded that planets are at least as abundant stars, and likely much more abundant, putting their number in the order of hundreds of billions to tens of trillions in our galaxy alone [Cassan et al., 2012].

Though unavoidable detection biases have skewed the dataset somewhat, it is today possible to obtain reasonable quantitative estimates of our galaxy’s planetary demography. The result has been a complete overturning of traditional planetary formation theory [Ford, 2014]. Far from being a universe dominated by cumbersome giants, it is now apparent that small rocky or icy planets in fact predominate [Han et al., 2014]. Furthermore, the most common class of planets appears to comprise those which are larger in radius than Earth but smaller than Neptune [Batalha et al., 2013]. A good example of this very popular new planet class is the system of three temperate, watery sub-Neptunes recently discovered in orbit around a nearby M-dwarf [Günther et al., 2019]. Our local solar system has no representative of this population and it has been speculated that, if one did exist, it may have been destroyed [Martin and Livio, 2016] or ejected [Cloutier et al., 2015] at some point long ago. These exciting new worlds are often close-orbiting and are tantalising candidates for life [Tsiaras et al., 2019].

Determining the interior structures, or even the density, of exoplanets discovered using radius-based methods can be difficult. Very large-radius bodies may be assumed to be Jupiter-like gas giants purely by the requirement that, if they were any denser, they would collapse. The density of smaller planets is more indeterminate; although there is some direct evidence from transit spectroscopy [Valencia et al., 2013] and ideally more in the near future [Beichman et al., 2016], typically we are forced to rely on a combination of statistical inference and broad mass-radius scaling [Rogers and Seager, 2009, Dorn et al., 2017b]. Such methods suggest that planets within the highly populous 1 - 4 Earth radius interval fall into two broad camps: ‘super-Earths’ which are rocky but potentially much more massive than our planet, and the smallest of the sub-Neptunes, implicitly volatile-rich planets

with only slightly greater masses but significantly larger radii [Dorn et al., 2017a]. Between these camps exists an apparent ‘radius gap’ between 1.5 to 2 Earth radii [Weiss and Marcy, 2014]; this has been taken to imply that an evolutionary tipping point exists above which runaway gas accumulation invariably turns rocky planets into icy ones. Modelling suggests the bifurcation is not a sharp one; some oversized rocky worlds and undersized gassy ones are expected [Wolfgang and Lopez, 2015]. Nevertheless, the ‘Neptune transition’ implies that the scope of ‘Earth-like’ planetary science can practically be neatly constrained to the range of half to double Earth-radius. Fortunately, there is a great abundance of such planets: thousands, in fact, within 50 lightyears alone [HEASARC, 2017, NASA Exoplanet Science Institute, 2019].

Perhaps because we have no local examples, super-Earths have invited particularly intense speculation. Much debate surrounds the question of whether such worlds should be expected to have plate tectonics or not. The debate is pertinent to purely Earth-based studies because it goes to the question of what plate tectonics actually requires in order to commence and continue. Indeed, the super-Earth scholarship is almost evenly split between those who claim plate tectonics is more likely on larger worlds [Valencia et al., 2007, Valencia and O’Connell, 2009, Korenaga, 2010, van Heck and Tackley, 2011, Foley et al., 2012], those who claim it is less likely [O’Neill and Lenardic, 2007, O’Rourke and Korenaga, 2012, Stamenković et al., 2012, Stein et al., 2013], and those again who argue that size has an inherently ambiguous effect [Karato, 2014, Stamenković and Seager, 2016, Kameyama and Yamamoto, 2018]. The disagreement does not break down simply along methodological lines, with numerical modelling, analytical arguments, and geochemical considerations variously employed in support of all three contentions.

The super-Earth debate arguably generates more heat than light, but it is nonetheless illuminating in that it reveals the true uncertainties of geodynamics where we are unable to rely on the arrogance of fact, as we can on Earth. It prompts consideration of whether we would even have a notion of plate tectonics if we did not observe it in action at home, and invites speculation about what completely alien solutions may exist to the fundamental question of planets.

2.4 Geodynamics: an evolving discipline

Tectonics, from the Greek *tectonicus*, ‘pertaining to building’, is a body of theory which endeavours to characterise the manner in which the spatial configuration of the landmasses changes over time. It is one of the foundational theories of natural history and arguably comprises the backbone of modern geosciences. However, tectonic theory is not, per se, a ‘dynamic’ theory with the power to explain how, or indeed, why such reconfiguration occurs. It is in this distinction that the field of geodynamics arises.

Geodynamics is to the theory of plate tectonics as genetics is to the theory of evolution. It seeks to interface the enormously successful narrative power of tectonics with the explanatory and predictive power of the underlying laws of physics - that is, classical mechanics and thermodynamics. Geodynamics defined in this way is necessarily younger than tectonic theory, which is itself only two generations old; as a discipline, it is arguably yet to move beyond descriptive science and acquire true predictive and prescriptive powers.

2.4.1 Empirical foundations of geodynamic thought

The world that geodynamics sets out to describe is characterised by several axiomatic properties. Some have been apparent since ancient times; others only much more recently. What is important to note is how powerfully suggestive they are of a unifying underlying phenomenon when assembled side by side, as follows:

- The Earth’s upper crust is partitioned chemically between felsic (continental) and mafic (oceanic) terrains [Lutgens and Tarbuck, 2012].
- The Earth’s crust is partitioned texturally between extensive smooth regions (plains and plateaux) and localised, predominantly arcuate rough regions (hills and mountains) [Van Der Linden, 1985].

- The Earth’s surface is partitioned topographically between dominant low-elevation regions (ocean floors and continental shelves) and scattered high-elevation regions (continental cores) [De Pater and Lissauer, 2015].
- The Earth’s upper crust is partitioned mechanically between broad areas of low or moderate strain rate (plate interiors) and narrow, globally interlaced bands of high strain rate (plate boundaries) [Kearey et al., 2009].
- The Earth’s lithosphere can be shown to be clearly partitioned into zones of common velocity when the vectors are constructed around Euler poles [De Pater and Lissauer, 2015].
- The Earth’s surface is partitioned chronologically between temporally heterogeneous zones of intermixed extremely young and extremely old rocks (continents) and temporally gradated zones made of rocks no older than 300 million years (ocean floors) [Coltice et al., 2012].
- The landmasses of the Earth are variously scattered into fragments or consolidated into supercontinents throughout deep time [Wegener, 1924].
- The bulk Earth is partitioned chemically and gravitationally between volatiles (atmosphere and hydrosphere), silicates (crust and mantle), and heavy metals (the core) [Lodders and Fegley, 1998].
- The bulk Earth is heated basally by conduction across the core-mantle boundary as well as volumetrically (although not necessarily homogeneously) by radiogenic decay [Turcotte and Schubert, 2014].
- The Earth, it is fairly well established, is round [Aristotle and Guthrie, 1939].

Such a bill of facts, once collected, testifies strongly to the existence of a global assortative process with both toroidal and poloidal components. Because of the apparent thermal and compositional gradients involved, the immediate sensible process thus implicated is convection, i.e. transport by thermal buoyancy. Convection implies zones of upwelling and downwelling: rheological features which can be identified geologically with ocean ridges [Macdonald, 1998] and trenches [Arculus et al., 2019] respectively. Better still, the pattern of geothermal fluxes aligns almost exactly with what would be expected if Earth’s surface plates represented the upper limbs of mantle-scale convection cells [Bennett et al., 2008]. All these virtues explain why, from the mid-20th century onward, the model of global plate movement driven by mantle convection has served as a unifying explanatory paradigm for all geoscience.

2.4.2 Challenges to the conventional model of plate tectonics

While the essential logic of convection-driven plate motion as a model for the Earth is sufficiently robust as to be beyond serious doubt, the scope of uncertainty at finer levels is daunting.

The Naive Theory of plate tectonics To illustrate how our understanding of global geodynamics has evolved, it is helpful to clearly articulate a ‘Naive Theory’ of plate tectonics to serve as a null hypothesis for the discussion. According to the Naive Theory:

- Earth is in a state of whole-mantle convection driven mainly by accretionary heat escaping from the core.
- Material and thermal heterogeneities, while certainly present, do not determine the global convective planform.
- Global convection is powered by the thermal gradient between the deep Earth and outer space, and should be expected to dwindle as this heat reservoir is depleted. Hence, we would expect elevated global geothermal fluxes in the past relative to the present.
- Plate motions represent the upper limb of these whole-mantle cells, with the plates themselves in some sense propelled by the overall mantle current. Implicitly, poloidal motions are diagnostic, while toroidal motion is merely a waste of convective energy.
- Mantle-scale upwellings should be expected to map to mid-ocean ridges and downwellings to subduction zones. Implicitly, plate scale forces such as ridge push, basal traction, and slab pull should cooperate to transport the plates according to whole-mantle circulation, and ought to be of similar magnitudes.

In general, the ‘Naive Theory’ is how the solid Earth would be expected to behave if it was dominantly basally heated, subject to linear viscosity laws, and chemically homogeneous to a first-order estimation. Such a rheology is powerfully amenable to numerical analysis and has been extensively studied as a common end-member behaviour for all sorts of systems in a state of thermally-driven flow (this thesis treats with similar rheologies in Chapter 3 and Chapter 4). Indeed, the ‘Naive Theory’ serves as an excellent approximation for systems ranging from Earth’s atmosphere to Campbell’s tomato soup.

Insufficiencies of the Naive Theory However, when applied to the bulk Earth, the Naive Theory fails in nearly every respect, even as a generalisation. For one, it massively overestimates the contribution of ridge push to plate motions: slab pull is acknowledged as by far the dominant factor [Forsyth and Uyeda, 1975]. It also understates the apparently very substantial role played by chemical heterogeneities, most obviously vis a vis continental crust and the supercontinent cycle [O’Neill et al., 2009, Zhong et al., 2007], but also, increasingly, in respect to the hypothesised basal chemical heterogeneities at the core-mantle boundary [Zhong and Liu, 2016, Torsvik et al., 2006], which some suspect to be integral to the global convective planform [Conrad et al., 2013, Arnould et al., 2018].

More generally, upwellings and downwellings do not always correspond with plate boundaries, are not in general symmetric, and can range in size from mantle-scale to lithospheric scales. With respect to downwellings, these can be plane-symmetric and aligned with (collisional) plate boundaries, as seen in subduction zones [Zhao, 2009], but may or may not be mantle scale depending on plate strength and upper mantle temperatures [Tanimoto and Lay, 2000, Čížková and Bina, 2019]; there is also a significant role played by delamination and dripping, which can occur on a range of scales and almost always remote from plate boundaries [Beall et al., 2017]. Upwellings, too, can be both mantle-scale and smaller-scale. ‘Hotspots’, as seen beneath Hawaii and East Africa, may fall into either category, depending on the interpretation of the tomographic and geochemical evidence. When major upwelling zones coincide with plate boundaries, it is strictly in the form of plumes, as seen in Iceland and East Africa [Lithgow-Bertelloni and Silver, 1998], with any planiform upwelling beneath the mid-ocean ridge being a strictly plate-instigated, shallow phenomenon [MELT, 1998]. Finally, toroidal motions are far from superfluous, but appear to actively enhance convective vigour by reducing viscous dissipation at the surface [Bercovici, 1993].

Naive Theory and the Urey Paradox One sense in which the Naive Theory is surprisingly apt is in its minimisation of the role of radiogenic heating relative to secular cooling as a driver of convection in the mantle. The proportion of surface heat flux accounted for by radiogenic decay in the mantle is called the Urey Ratio [Turcotte and Schubert, 2014] and it was originally supposed to be close to unity: that is, the Earth releases its heat as rapidly as it is produced [Urey, 1955]. This assumption was required to circumvent the so-called ‘Urey paradox’, in which unrealistic temperatures are obtained in the early Earth if today’s geothermal flux is mostly accounted for by secular cooling. However, multiple lines of evidence - including estimates of Earth’s original radiogenic inventory based on chondritic compositions [Korenaga, 2003], as well as state-of-the-art geoneutrino surveys [Gando et al., 2011, Mareschal et al., 2012, Huang et al., 2013b, Machulin and Collaboration, 2015] - now agree on a much lower Urey ratio, somewhere in the neighbourhood of one third [Mareschal et al., 2012, Korenaga, 2011].

In one sense, a low Urey ratio is favourable for the Naive Theory, as it implies that basal heating dominates volumetric heating: a state of affairs that promotes vigorous long-wavelength convection [Weller and Lenardic, 2016, Wolstencroft et al., 2009]. However, to avoid the Urey paradox, this would require an inverse relationship between global geothermal flux and net mantle temperatures - in other words, it seems that the Earth sheds its heat more rapidly the cooler it gets [Korenaga, 2003]. This counter-intuitive behaviour is hard to evoke with the kinds of isoviscous or exponentially temperature-dependent rheologies that the Naive Theory supposes.

Going beyond the Naive Theory We have employed the Naive Theory here mostly as a discursive device. However, though no theory of this form has ever been seriously advocated by qualified

geodynamicists, the Naive Theory is a surprisingly close representation of common opinion regarding solid Earth dynamics, both among the lay public and in the non-specialist planetary and geoscientific literature. This can be seen, for instance, in the generally accepted supposition that Mars and the Moon lack active tectonics due to their lesser size, and hence - presumably - their more rapid rate of cooling. The resilience of Naive-like assumptions about the Earth is likely due in some degree to the fact that, as misguided as the theory is, its elegant formulation and superficially potent explanatory qualities still recommend it over alternative models, the underlying causal mechanisms of which remain elusive.

So, what might an improved, unified geodynamic model of the Earth entail? We can draw up a decent catalogue of what such a model would require by merely inverting the assumptions of our by now well-falsified null hypothesis:

- Earth is in a state of mixed convection on multiple interconnected wavelengths, with the dominant driver being negative buoyancy of the lithosphere.
- Chemical and thermal heterogeneities are significant and have a first-order effect on global convective planform.
- Instead of convective vigour being determined by global temperatures, temperatures are in fact determined by convective vigour, which has a range of complicated dependencies. Global heat flux can go inversely with global temperature and potentially exhibit many other counter-intuitive feedbacks.
- Plate distributions powerfully condition global flow, and are themselves a function more of historical and local contingencies than of deep perturbations.
- Mantle-driven upwellings and downwellings are to some degree separate from surface-driven movements, with the two systems at times reinforcing and at times disrupting one another.

What conceivable system, governed by what conceivable laws, could provide for these behaviours? Over the past two decades, significant progress has been made from various quarters to provide a new paradigm that satisfies these requirements without stipulating causes beyond the essentials of thermally driven flow. Two aspects of this paradigm will now be discussed: the discovery of tectonic modes, and the special role of continents.

2.4.3 Continents: a decisive control in the tectonic system

Much of the progress of recent years is owed to the development of robust and versatile computational numerical modelling tools, which have proved to be particularly efficacious for investigating non-linear, and indeed, highly counter-intuitive systems [Moresi et al., 2007]. The story they tell is one of a global flow regime controlled by chaotic interactions between high Rayleigh number instabilities and pronounced material heterogeneities: in particular, the strong, cold, low-density materials that make up the continents.

Thermal effects of continental crust One major observation from recent geodynamic modelling is that a planet's global flow regime needn't necessarily be that which maximises heat flux [Grigné et al., 2005]. A good example is the effect of continental crust on mantle temperatures. Intuition holds that the presence of thick continental lithosphere should insulate the mantle, leading to reduced net surface heat flux [Korenaga, 2008, Gurnis, 1988, Anderson, 1982]. This feedback is often given a role to play in the periodic assembly and breakup of supercontinents, by triggering the formation of large sub-continental mantle plumes [Brandl et al., 2013, Yoshida and Santosh, 2018].

Modelling, however, no longer supports this. Even in purely thermal models, supercontinentality does not necessarily lead to increased sub-continental temperatures [Heron and Lowman, 2011]. In dynamic models, only fairly sluggish rheologies experience any meaningful sub-continental warming from insulation [Heron and Lowman, 2014, O'Neill et al., 2009]; when geochemistry is considered, the effect of the partitioning of heat-producing elements into continents can be shown to substantially outweigh any insulation effect [Cooper et al., 2006]. In fact, under more sophisticated rheologies, increased vigour due to transient warming [Lenardic et al., 2005] and geometric flow conditioning imposed by supercontinentality [Zhong and Liu, 2016] may in fact boost the global surface flux. In this

way, the apparent regularity of the supercontinent cycle - the first-order feature of Earth's Phanerozoic history - could be accounted for purely by surface-driven forcings of the underlying mantle, without the need for catastrophic interventions like massive mantle plumes.

Continents as a resolution of the Urey paradox Continents may also provide a partial resolution to the Urey paradox. With some exceptions, estimations of the global heat budget over time have not explicitly taken into account the role of continents [Conrad and Hager, 2001, Korenaga, 2008]. Numerical modelling suggests that mantle heat flow, to a first order approximation, is unaffected by historically typical degrees of continental coverage [Lenardic et al., 2011]. As higher continental abundances imply lesser radiogenic forcing in the mantle due to partitioning, the net result is a lower Urey ratio for much of Earth's history. Feedbacks of this sort resolve the Urey paradox without the need for untestable heterogeneous mantle models [Korenaga, 2011].

Continent-climate feedbacks Another under-appreciated role of continents in mantle convection comes indirectly by way of the atmosphere. Through albedo [Hyde et al., 1990], silicate weathering [Raymo and Ruddiman, 1992, Berner and Berner, 1997], carbon storage [Marshall et al., 1988], and topographic effects [Birchfield and Wertman, 1983, Stolar et al., 2006], continents have long been acknowledged as a potent factor in long-term global climate.

However, it is only recently that climate forcings on mantle flow have begun to be recognised. Numerical suite modelling shows that mantle flow can be suppressed as surface temperature rises [Weller et al., 2015]; conversely, modelling of subduction zones suggests the opposite coupling, as global warming enhances sediment runoff, lubricating subduction and possibly accelerating mantle overturn [Behr and Becker, 2018, Sobolev and Brown, 2019]. The net effect of climate may be so profound as to induce a complete switch in the style of tectonics [Lenardic et al., 2008, 2016b]; the picture only becomes more complicated when the variegated feedbacks between continental drift, evolution, and bioclimatic controls are also taken into account [Zerkle, 2018]. Whether climate is a stabilising or destabilising agent in solid Earth circulation remains in question, as does the significance of tectonic-climate feedbacks to the maintenance of planetary habitability [Foley, 2015].

2.4.4 Tectonic modes: beyond plate tectonics

In the 1990s, the failure of linear mantle models to capture the major features of plate tectonics, even at extreme Rayleigh values [Moresi and Solomatov, 1995], motivated the development of non-linear, stress-dependent rheologies [Bercovici, 1993, Solomatov, 1995]. These models had the virtue of exhibiting strain-localisation, wherein the upper boundary layer separates into discrete units that can be generalised as plates [Bercovici, 1995, Zhong et al., 1998]. The new models supported many interesting discoveries; most notably for our purposes, the discovery of the first non-trivial 'tectonic mode' to be observed outside of nature, the 'episodic overturn' mode [Moresi and Solomatov, 1998], now widely believed to be operative on Venus [Turcotte et al., 1999, Huang et al., 2013a]. The discovery of episodic overturn gave credence to the importance of 'phases' in rheological parameter space; as applied to planets, these phases became known as 'tectonic modes' and are today a major topic of study. While some are fairly trivial, like the 'stagnant' regime, others are more surprising.

Mobile lid When the convective vigour is much higher than the yield stress, a state is obtained where part or all of the lithosphere is failing at any given time. The 'mobile lid' category as it is presently defined is generous: it encompasses plate tectonics as well modes exhibiting very broad, or even uniform, lithospheric strains patterns.

Under an isoviscous rheology, a mobile tectonic mode persists in all but the most extreme cases [Blankenbach et al., 1989]; nevertheless, there are subtleties to the mobile lid that are presented in depth in Chapter 3.

Stagnant lid When the yield stress is very high compared to the convective vigour of the mantle, a very strong, thick, static lithosphere may be formed. In a continuum mechanics sense, a stagnant lid

could be characterised as any flow regime in which the maximum strain rate at the upper boundary layer is close to zero. More recently, authors have distinguished between a hot stagnant lid, as seen perhaps on the very early Earth [Debaile et al., 2013, O'Neill and Debaile, 2014], and a cold stagnant lid, represented by the present-day Moon [Weller et al., 2015].

The stagnant lid mode is a typical consequence of depth- or temperature-dependent rheologies such as Arrhenius-type viscosity [Davaille and Jaupart, 1993], which is the partial subject of Chapter 3.

Episodic overturn When the yield stress is nominal relative to convective vigour, a temporally unstable regime ensues in which long periods of lid stagnation are interspersed with abrupt failure events. Episodic overturn has a special place in tectonic mode theory: in models that reproduce a brittle lithosphere [Moresi and Solomatov, 1998] in which a parameter implementing mechanical strength is the control variable, the episodic regime can be shown to occupy an intermediary position between stagnant and mobile lids. This observation demonstrates the feasibility of a ‘phase diagram’ type conception of planetary tectonic mode: testing the applicability of this paradigm is an ongoing and active field of research [Driscoll and Bercovici, 2013, Weller et al., 2015]. Venus is commonly considered a representative of this class based on the uniformity of its surface ages, around 300 My [Turcotte et al., 1999].

Episodic overturn is the main subject of Chapter 4 of this thesis.

Magma ocean In a magma ocean, the upper lithosphere is by definition in a state of global, continuous, contiguous failure: this is the definition of a fluid. Most magma ocean models have only a shallow ocean, and most are short-lived - indeed, so short as to be below the characteristic thermal timescale [Andrault et al., 2011]. Hence, characterising the magma ocean as a tectonic mode is somewhat fraught. The Hadean eon of the Earth is so-called because of the assumption that a sustained magma ocean existed at this time [Abe, 1993], and it is traditionally assumed that this is so for all the rocky planets [Schaefer and Elkins-Tanton, 2018], though mounting evidence of a ‘cool early Earth’ increasingly call this into question [Valley, 2005, Kenny et al., 2016]. The Moon and other small rocky bodies preserve the most unambiguous evidence of magma ocean tectonics, with the Moon’s highland-lowland dichotomy serving as the classic fossil example [Warren, 1985, Elkins Tanton et al., 2002, Elardo et al., 2011, Elkins-Tanton et al., 2011, Ogawa, 2018].

The magma ocean, and the manner and timing of its demise, is an important consideration for models of the early Earth (Chapters 5 & 6).

Sluggish lid If viscosity is only moderately temperature-dependent, a ‘sluggish’ regime can develop [Solomatov, 1995], with large aspect-ratio convection cells leading to diffuse deformation zones in the upper boundary layer. Although the lid is technically mobile in this scenario, the motion is driven by mantle currents rather than vice versa, resulting in relatively tepid convection [Bercovici et al., 2000] and limited or strictly sub-lithospheric recycling [Beall et al., 2017]. While Earth is sometimes considered to have passed through a sluggish lid phase [Foley, 2018], Mars is the oft-cited bellwether for this mode, based on its extensive deformed but ancient terrains [Banerdt et al., 1992], isotopic analysis of Martian meteorites [Debaile et al., 2009], and numerical modelling [Kiefer, 2003]. A sluggish lid can be recognised by the existence of substantial toroidal flow in the upper boundary layer in the absence of significant poloidal flow.

The possibility of a sluggish lid on the early Earth is contemplated in Chapters 5 and 6 of this thesis.

Plume tectonics If the defining characteristics of plate tectonics are horizontal motion, arcuate deformation, and surface-driven convection, plume tectonics represents the opposite: vertical motion, bulbous deformation, and basally-driven convection. Plume tectonics, also called ‘vertical tectonics’, preceded plate tectonics as the popular explanation for the global distribution and structure of major landforms [Van Bemmelen, 1976], and it remains popular with those unconvinced by a dominantly surface-driven global circulation regime [Maruyama, 1994, Kumazawa and Maruyama, 1994, Tsuchiya

et al., 2013]. The picture is complicated by the fact that the two substantially independent systems [Hill et al., 1992] nevertheless seem to not only co-exist but cooperate on the Earth, with plumes inducing, or being induced by, the formation of triple junctions at the surface [Burke and Dewey, 1973], a process which apparently goes back to the Archaean [Van Kranendonk, 2010]. Plume tectonics is also connected with plate tectonics in that it is potentially the source for much of Earth’s early continental crust [Hill et al., 1991, Walzer and Hendel, 2017] via a process which may be ongoing today; consider the oceanic plateaux, such as the Kerguelen and Azores, which arguably represent Phanerozoic protocontinents [Albarede, 1998, Condie, 2000, Aulbach, 2012]. It has lately been hypothesised that plume tectonics not only preceded plate tectonics on the Earth [Fischer and Gerya, 2016] but actively gave rise to it [Gerya et al., 2015], by means of any one of numerous modelled mechanisms [Mason et al., 2010, Rey et al., 2014].

Controversies aside, plume or vertical tectonics is a vital part of the emerging tectonic mode paradigm. Beyond Earth, plume tectonics is the preferred explanation for the Tharsis terrain on Mars [Mège and Masson, 1996, Ernst and Buchan, 2001, Baker et al., 2007, Dohm et al., 2007, Hynes et al., 2011], where it appears to have been localised, massive, and extremely long-lived [Werner, 2009]; it is also implicated by various features on the surface of Venus including coronae and nova [Schubert et al., 1989, Stofan et al., 1992, Smrekar and Stofan, 1997, Johnson and Richards, 2003, Ernst and Desnoyers, 2004, Stofan and Smrekar, 2005].

The relationship between plume tectonics and plate tectonics, and the possibility and mechanisms of a plume-to-plate transition, are the main subject of Chapters 5 and 6 of this thesis.

Heat pipe The heat pipe model was first postulated to explain certain observations of Jupiter’s moon Io [O’Reilly and Davies, 1981], but it wasn’t until more recently that it attained a rigorous thermodynamic treatment [Moore, 2001]. Today it is recognised as a distinct tectonic mode - arguably the second truly non-trivial mode to be discovered after episodic overturn [Stevenson, 2003]. In this model, a very strong, thick, cold lithosphere overlies a hot interior; long-lived plumes generate persistent volcanism on the surface, which continually re-buries the vacuum-chilled planet surface and so cools the interior by a process of pressure-driven downward cold advection [Moore et al., 2017]. Heat pipe tectonics is even more efficient at discharging internal heat than plate tectonics - Io’s surface heat flux is estimated at an incredible 40 times that of Earth [Veeder et al., 2012].

The heat pipe mode is one resolution to the problematic jigsaw puzzle of the early Earth heat budget [Moore and Webb, 2013, Griffin et al., 2014]; though it has yet to be convincingly replicated in numerical modelling, it remains a hot topic in the literature.

Null mode In the interest of completeness, the case of a planet thoroughly at equilibrium with the temperature of free space should be mooted. Such a mode, here called the ‘null mode’, is a perfect attractor in parameter space upon which all planets must eventually converge if left to their own devices. Dwarf planets such as Ceres likely approximate the null mode in our solar system; if the Earth were set adrift in interstellar space, it would reach such a state on the order of 100 Gy from now [De Pater and Lissauer, 2015].

The future of tectonic modes The discovery of the episodic overturn and heat pipe modes makes it almost certain that more unseen modes remain to be uncovered. While limitless novelty can be generated with more and more complex material configurations, the real test of the paradigm will be to show that single-material, mathematically parsimonious laws, like the viscoplastic formulation that enables episodic behaviour, can continue to generate substantially original behaviours that can be called truly new tectonic modes.

2.4.5 Tectonic mode transitions and planetary memory

So far, we have encountered mode transitions as diagnostic or taxonomic classifications of viable systems for processing geothermal heat. This is a geophysical analogue of how weather and climate may be understood fundamentally as systems for processing solar heat. According to this manner of

thinking, planets should be expected to ‘select’ the most efficient system for processing their heat that is made possible by their circumstances at any given time [van Thienen et al., 2005].

We have already seen that a planet’s geothermal flux must not be naively expected to decrease as a function of time, or even as a function of temperature: both may indeed have the opposite effect. However, that does not invalidate the null hypothesis that the flux should be the highest possible.

Consider a schematic tectonic history of the Earth [Stern, 2018]: first, a magma ocean blasted by accretionary heat, core separation, and short-lived radiogenics; then, as surface cooling drove the magma to depth, an Io-like heat-pipe phase; third, a sluggish plume-tectonic phase akin to Venus or perhaps Enceladus; and finally, with sufficient cooling yielding a sufficiently brittle lithosphere, the breakup of one plate into multiple plates and the dawn of plate tectonics. According to this history, geothermal flux starts high when global temperatures are high, dips through the single-plate era, then rises with the advent of subduction; at each stage, the tectonic mode is a direct and unitary consequence of the state variables of global relict and radiogenic heat. This is what we will call the ‘phase diagram’ model of tectonic mode transitions, wherein a planet tracks a path across a multidimensional tectonic phase space [Sleep, 2000, O’Neill et al., 2007, Sleep and Schubert, 2007] just as a block of water ice may be compelled by its state variables of temperature and pressure to cross the liquid domain on its way to being vapour.

This way of viewing tectonic modes is increasingly challenged by geodynamic numerical modelling. A number of groups have recently furnished evidence that a planet’s choice of tectonic mode is inherently multistable [Pla et al., 2009, Weller et al., 2015, Lenardic et al., 2016a], indeterminate [Lenardic and Crowley, 2012], remote from equilibrium [Weller et al., 2015], and heavily influenced by stochastic [Lenardic et al., 2016b], contingent [O’Neill et al., 2015, 2016], historical [Weller and Lenardic, 2012], and circumstantial factors [Glukhovskii and Kuz’mín, 2015, O’Neill et al., 2017, O’Neill et al., 2018].

These lines of evidence suggest a new model: one in which tectonic mode is treated not as a state variable or a mere product of state variables, but rather as a process variable - a configuration that flows through the system, like heat itself [Lenardic, 2018]. Under such a model, multiple tectonic modes could be seen as coequal simultaneous possible configurations for each planet, each manifesting certain exigencies and instabilities that could spontaneously and unpredictably precipitate a transition into an alternative mode. Thus two planets identical in every way may, in sufficient time, evolve to be more and more unlike, until the ultimate logic of heating and cooling forces both to converge on stagnation and eventual, frigid death.

Though the new model is not yet well quantified, there are many facts we have already seen that should predispose us to suspect it may be true. Consider again the analogy with weather and climate. Non-linearities and chaotic behaviour are as influential there as it is here claimed they are in the solid Earth. However, there are two key differences. Firstly, planetary tectonic mode not only processes its heat source, but feeds back into it by determining the rate at which it is conveyed to space. On Earth, we see evidence of this effect most definitively in the lower-than-expected modern Urey ratio, which implies that Phanerozoic surface processes have in fact created a much more efficient channel for extracting heat from the core than existed prior. Secondly, planets have long-term memories. While the oldest climate information with direct relevance to the modern atmosphere is ~200 My old carbon, Earth’s tectonic processes can be affected by circumstances billions of years old: for instance, when a buoyant Archaean craton impacts a subduction zone and forces a reconfiguration of plate stresses or even plate boundaries [Mason et al., 2010].

While ultimately it may be tempting to cut through the uncertainty and hysteresis and return to the basic intuition that the Earth is becoming thermally ‘older’ with time, with less net heat to go around and hence less prospect for activity, it must again be stressed that the actual rates of flux, even in our elevated present regime, are very small with respect to tectonic timescales. Even a seemingly irreversible process like the formation of cratons on Earth is conceptually not undoable: as we calculated earlier, sufficient potential energy remains inside the Earth to effect the ‘rewinding’ of that particular degree of freedom. In principle, it might even be possible to establish a circle - or, more properly, a very shallow spiral - of alternating tectonic modes, which, if sufficiently regular, might be construable as a distinctive tectonic mode of its own: consider episodic overturn [Moresi and Solomatov, 1998]. Such a cycle, despite the inevitable ratcheting of entropy, could nonetheless

conceivably be stable even on timescales comparable to the present age of our universe.

2.5 Review

The emerging model of tectonic modes will demand novel and imaginative approaches. This study's particular methodology will be discussed forthwith (Chapter 2); for now, we must conclude with a reaffirmation of the essential thesis. The age of planets as dumb rocks mechanistically unspooling into space has come to an end. Felled by a torrent of empirical evidence at home and far beyond, that conception is now being replaced with a new one: broader, richer, more ripe with possibilities than what came before; but also challenging, subtle, and bewilderingly vast. It is to be hoped that this new paradigm will in time yield new tools; tools which, if wielded with boldness and ingenuity, may open up the tomb of Earth's ancient past, and lay bare the pathways of life.

Chapter 3

Foundations

Under the operating paradigm of our research program, the phenomena of tectonics as we know them are held to be the surface expressions of a global system of solid-state convection. Tectonics - literally the ‘construction’ of Earth and other planets’ surface geology on a global scale - will thus be studied by proxy of the mantle-scale circulations which are inferred to motivate it. Because the theory justifying this substitution mandates certain controls, assumptions, and simplifications which must be accommodated by our modelling methodology, we will discuss these matters a little here.

3.1 The analytical toolkit

Tectonics is known to us through its sensible processes of orogeny, seismicity, and volcanism. The energy available to carry out these permutations ultimately derives from the depletion of the thermal gradient of the Earth’s hot interior with space, mitigated to an uncertain degree by internal heat production via radiogenics, core despinning, and other means. Estimates of global heat flow vary from around 42 terawatts [Dye, 2012] to upwards of 47 terawatts [Davies and Davies, 2010]. Of this power, a mere 1% is thought to be necessary to account for all the geological activity witnessed on Earth [Turcotte and Schubert, 2014]; if our Earth is a heat engine, it is a weak one.

The essential model of planetary solid-state thermal circulation is Rayleigh-Benard convection, in which a fluid held between two plane layers of different temperatures is observed to spontaneously self-organise into counter-rotating cells to maximise the efficiency of transport [Getling, 1998]. Characterising the geometry or ‘planforms’ of convection, and finding what determines them, is the core business of this thesis.

3.1.1 The Nusselt number

A geodynamically rigid planet with Earth’s interior temperature would not be able to access even these modest energies: it would be trapped by its flat, linear conductive geotherm. That the planetary geotherm is evidently much greater than this is evidence that more kinetic processes are at work. The dimensionless temperature gradient is related to the Nusselt number or Nu , the ratio of the measured temperature gradient to the reference gradient, which is the purely conductive geotherm. It can be given in terms of the rate of change of the dimensionless potential temperature θ^* with respect to dimensionless height from mantle base h [Schubert et al., 2001]:

$$Nu = - \left| \frac{\partial \theta^*}{\partial h} \right|_S \quad (3.1)$$

Where $|x|_S$ indicates the average value across a surface. (We could alternatively have used the depth z instead of the h : it has no impact on the dynamics.) The asterisks indicate a non-dimensionalised quantity: this is a convention throughout the literature. When dimensionless parameters are used - unit mantle thickness and unit temperature range - the conductive geotherm for a non-curved domain is exactly one. In non-curved domains, Nu is equivalent to the dimensionless surface temperature gradient, and so it is confusingly defined as such in some contexts [Blankenbach

et al., 1989]. For curved domains, where the outer length is greater than the inner length, the conductive geotherm is proportionately lesser as it is in a sense ‘stretched out’ across the circumference; letting f be the ratio of inner to outer lengths (either circumferential or radial), Nu in these cases diverges from the dimensionless temperature gradient by a factor of f for cylinders and f^2 for shells.

There is a practice in some quarters of adding a constant 1 to Nu, reflecting a difference of opinion over whether Nu is best constructed as an arithmetic quantity (i.e Nu as the convective flux after conductive flux is subtracted) or as a geometric quantity, as we have stated it here. We prefer the latter usage, reterming the former as Nu_+ .

Though harder to measure in practice than in theory, it is implicit that Earth’s Nusselt number must be much greater than one; it is sometimes cited in the order of 10 [Tackley, 1996], which is characteristic of laminar (sub-turbulent) flow [White, 1984].

3.1.2 The Prandtl, Grashof, Reynolds, and Rayleigh numbers

If conduction is insufficient to explain Earth’s geotherm, another process is implicated, and that is free convection - buoyancy-driven advection of heat. The relative effectiveness of convection is a product of three further dimensionless quantities. The *Prandtl* number Pr is a ratio of *diffusivities* - quantities in units of area per time; in this case, the ‘momentum diffusivity’ or *kinematic viscosity* ν and the thermal diffusivity κ :

$$\text{Pr} \equiv \frac{\nu}{\kappa} \quad (3.2)$$

$$\nu = \frac{\mu}{\rho} \quad \kappa = \frac{k}{\rho c_p} \quad (3.3)$$

Where μ is the dynamic viscosity, ρ is density, k is thermal conductivity, and c_p is specific heat. The *Grashof* number Gr, meanwhile, concerns the forces involved: it is the ratio of buoyancy to viscous drag:

$$\text{Gr} = \frac{g\beta\Delta TL^3}{\nu^2} \quad (3.4)$$

Where g is gravity, β is the coefficient of volume expansion, ΔT is the temperature drop, and L is a representative length scale. Without a sufficient *Prandtl* number, heat will escape from each parcel faster than the parcel itself can be transported by buoyancy, while a low *Grashof* number would imply that the drag of the medium on each parcel is too great for buoyancy to overcome.

The third import dimensionless quantity is the *Reynolds* number Re, the ratio of inertial forces to viscous forces and thus a measure of flow turbulence:

$$\text{Re} = \frac{uL}{\nu} \quad (3.5)$$

Where u is flow velocity and L is the length scale. High values of Re imply that inertial forces dominate over viscous forces and tend to produce what we both professionally and colloquially term ‘turbulent’ flow. Another way of looking at Re is that it stipulates whether the velocity field must be treated as a state variable, like the temperature field, or as a dependent variable.

In free (unforced or ‘natural’) convection, we find that the *Reynolds* and *Grashof* numbers are related:

$$\text{Gr} = \text{Re}^2 \quad (3.6)$$

The product of Gr and Pr (or equivalently, Gr and Re^2) produce a crucial dimensionless quantity for characterising thermal fluids: the *Rayleigh* number Ra. The *Rayleigh* number, often termed the ‘convective vigour’, can be interpreted as the ratio of the diffusive and convective time scales in the medium; i.e. Ra serves as the *Peclet* number for heat. A simple formulation is as follows:

$$\text{Ra} = \frac{\alpha g \Delta T b^3}{\kappa \nu} \quad (3.7)$$

For high values of Ra , convection is much more efficient than conduction for transporting heat, leading to high fluid velocities and flow regimes grading from sluggish to laminar to turbulent. For low Ra , conduction dominates, and the material is largely or totally quiescent. Separating these two domains is an often empirically-obtained value, the Critical Rayleigh Number or Ra_{cr} , which is innate to each fluid. Ra is sometimes given in terms of Ra_{cr} as:

$$\text{Ra}^* = \frac{\text{Ra}}{\text{Ra}_{\text{cr}}} \quad (3.8)$$

What occurs at any given value of Ra^* is determined by the availability of a certain ‘critical’ wavelength of thermal perturbation. A thermal perturbation at the critical wavelength is guaranteed to grow faster than a perturbation of any other wavelength. Whether this is fast enough to outstrip conduction is a function of Ra . At $\text{Ra}^* < 1$, the conductive timescale is always quicker than the convective timescale, even for a critical perturbation, making convection impossible. At values of $\text{Ra}^* > 1$, there is always at least one perturbation wavelength that is fast enough to outstrip conduction and establish a convective geotherm. Convection is thus possible, but entirely dependent on the presence of an appropriately shaped perturbation. Increasing Ra^* beyond 1 makes ever more wavelengths available for convective growth, until at extreme values ($\text{Ra} \gg 10^7$) even artificial heterogeneities introduced by numerical noise can grow. Convection in such systems is inevitable, and large-scale models are overwhelmingly time-dependent [Jarvis, 1984]. Between these sub- and super-critical scenarios lies the point $\text{Ra}^* = 1$, where the growth timescale of a critical perturbation is comparable to the conductive timescale, and the outcome for the system depends entirely on the presence, frequency, and amplitude of exactly the right kind of perturbation.

Values of Ra in most applications can be quite high, and so are usually represented in decimal orders of magnitude; for mantle materials as modelled hereafter, for example, the critical Ra can be shown to be somewhere between $10^3 - 10^4$, with ‘Earthlike’ behaviour scarcely manifest anywhere below 10^7 (Chapter 3). Although Ra_{cr} is often obtained empirically, it can be derived from first principles for certain simple cases, as will be shown.

The *Rayleigh* number is a powerful tool for interrogating the behaviours of convecting fluids; however, the correct parameterisation of such a heavily compound term is a nuanced affair. Several assumptions are commonly made in the context of mantle circulation which simplify matters at the cost of limiting the scope of validity.

The ‘infinite *Prandtl*’ assumption asserts that momentum diffusivity is incomparably greater than thermal diffusivity; i.e.:

$$\nu_r \gg \kappa_r \quad (3.9)$$

This is a defensible assumption for the Earth, where the estimated value of the Prandtl number is around 10^{23} [Schubert et al., 2001]. Implied by the above, but worth stating clearly, is that the *Reynolds* number of the system - the ratio of inertial to viscous forces - approaches zero: i.e. inertia is negligible, present velocity is independent of previous velocity, and turbulence is consequently impossible. This follows because the thermal *Peclet* number, which is Ra , must be a product of Pr and Re^2 ; hence, to be finite, its expression in terms of Pr and Re must cancel at the limit.

The infinite *Prandtl* statement is often taken in tandem with the *Boussinesq* approximation, which neutralises all density-driven force terms which are not coefficients of gravity; in other words, the fluid is held to be incompressible:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (3.10)$$

Where u and v connote horizontal and vertical velocity components respectively. The incompressibility assumption in two dimensions allows us to define a stream function $\Psi(x, y)$:

$$u = \frac{\partial \Psi}{\partial y}, \quad v = -\frac{\partial \Psi}{\partial x} \quad (3.11)$$

$$\bar{u} = \nabla \Psi \quad (3.12)$$

Where $\bar{u}$ is the velocity vector and ∇ is the familiar vector differential operator *nabla* or *del*.

The stream function has many useful properties: lines of constant Ψ are called streamlines and are everywhere parallel to the velocity vector at that point, and a difference in value between any two points defines the volumetric flux across a line connecting those points, or equivalently the advective flux when multiplied by density ρ . (The absolute value of Ψ , however, is arbitrary.)

In addition to the *Boussinesq* and infinite *Prandtl* assumptions, we may further assert that the gravity is always radial and varies only with depth, and also that the fluid is inelastic, i.e. it has no stress memory. Together these several approximations hold wherever a dense, viscous fluid is subject to extreme pressures over relatively large spatio-temporal scales; hence they are held to be broadly - though not unconditionally - appropriate for mantle problems.

With the aid of this toolkit of assumptions, together with the constitutive equations for conservation of mass and energy, it is possible to obtain velocity and pressure solutions for the otherwise insoluble *Navier-Stokes* equations: the conservation of momentum equations for viscous fluids. The derivation for mantle problems is canonical but lengthy; details can be found in the universally cited textbook literature on the topic [Schubert et al., 2001, Turcotte and Schubert, 2014]. One extremely useful product, however, is a family of robust parameterisations of the *Rayleigh* number for mantle convection.

There are many notation schemes for these very common equations, and here we have elected for one that more clearly aligns with our computational approach (discussed later). In it, we define two new terms: a dimensionless ‘scale coefficient’ ζ that gathers the spatiotemporalised factors, and a ‘temperature scale’ factor τ representing the thermal inputs to the system. The familiar thermal expansivity term α , with units of ‘proportional change per temperature’, non-dimensionalises τ however it is defined; the term $\alpha\tau$ could thus be thought of as a kind of ‘characteristic size variation’ for the system - the amount of deformation a given parcel of matter should expect to experience over the course of its time within the system. The factor τ itself is obtained in different ways depending on the heat sources within the system: the basal and volumetric (radiogenic) heating parameterisations are shown, where c_p is the specific heat capacity.

$$\begin{array}{ccccccc} \text{Rayleigh} & & \text{Scale} & & \text{Thermal} & & \text{Temperature} \\ \text{number} & & \text{coefficient} & & \text{expansivity} & & \text{scale} \\ \text{Ra} & = & \zeta & \bullet & \alpha & \bullet & \tau \\ & & \equiv \frac{gb^3}{\mu\kappa} & & & & = \begin{cases} \Delta T & (\text{basal}) \\ \frac{b^2}{\kappa c_p} H & (\text{volume}) \end{cases} \\ \text{(dimensionless)} & & \text{(dimensionless)} & & \text{(temperature}^{-1}\text{)} & & \text{(temperature)} \end{array} \quad (3.13)$$

Where b is the vertical length scale (the distance from top to **bottom**).

Because the *Rayleigh* number so expressed is equivalent to the coefficient of the buoyancy term, it should now be clear why it is often simply dubbed ‘convective vigour’, as that is its primary effect. By parameterising the system in this way, the behaviour of seemingly distinct scenarios can be seen to be related through their common *Rayleigh* number; what’s more, a dimensionless treatment of the problem can be readily converted to a dimensionalised one by expanding the terms of Ra with their empirical or inferred values.

3.1.3 Linear stability analysis and the critical *Rayleigh* number

Remarkably, the value of the critical wavelength Ra_{cr} is independent of the thermal properties of the system and, for the sorts of plane box geometries often under discussion, should theoretically be exactly $2\sqrt{2}$ - or just $\sqrt{2}$ in the half cell [Chandrasekhar, 1961]. This can be obtained empirically, but in simple cases such as that of planar basally-heated isoviscous flow, an expression for Ra_{cr} due

to arbitrary perturbations can be derived from the assumptions already held using linear stability analysis.

First consider the state of a purely conducting system at thermal equilibrium:

$$T_c^*(h) = \frac{T_o}{T_i - T_o} - h + 1 \quad (3.14)$$

Where T_c^* is the dimensionless conductive temperature at dimensionless depth h and T_o and T_i stand for the outer (upper) and inner (lower) temperature respectively. Put simply, there is a linear dependency of temperature and depth.

Let us now impose a thermal anomaly θ'^* , uncertain in wavelength and infinitesimal in amplitude:

$$\theta'^* \equiv T'^* - T_c^* \quad (3.15)$$

Where the starred notation indicates a non-dimensionalised parameter and the prime notation, here and hence, identifies a perturbation. The choice of θ here relates to potential temperature, the quantity conserved along adiabats, which is what this perturbation will ultimately induce.

Before perturbation, the pressure gradient forces were defined solely by the hydrostatic pressure p_c - the pressure field which is purely sufficient to counteract the force of gravity. After the introduction of the perturbation, but before the resultant perturbed state is realised, the pressure field is modified in two ways: by the buoyancy anomaly of the perturbation, but also by the contribution of the modified density of the parcel to pre-perturbative hydrostatic pressure. Taking this into account, we can define a true perturbation pressure Π'^* as:

$$\Pi'^* \equiv p'^* - p_c^* \quad (3.16)$$

Where p'^* is the pressure deviation relative to the hydrostatic pressure.

What determines if this seed of chaos shall grow? Equivalently, we may ask which is faster - the growth of the anomaly, or the ambient restoring forces. The answer depends in part on the wavelength of the perturbation and in part on the overall convective vigour of the system; a very lengthy expansion [Schubert et al., 2001] reaches the sixth derivative before delivering the following relation:

$$\text{Ra}_{\text{cr}} = \frac{\pi^4}{4\lambda^{*4}} \cdot (4 + \lambda^{*2})^3 \quad (3.17)$$

Where $\lambda^* = \frac{\lambda}{b}$, the wavelength of perturbation equivalent to the original anomaly θ'^* in the horizontal coordinate, expressed as a ratio of the layer thickness b , and Ra_{cr} is what we came for: the ‘critical’ *Rayleigh* number above which perturbations of a given wavelength will grow more rapidly than they are diffused. The expression defines a curve through the space of Ra_{cr} vs dimensionless wavenumber which has a single minimum: this is $\lambda^{*,\text{cr}}$, the wavelength of perturbation at which Ra_{cr} is at its lowest. Perturbations near this critical wavelength will tend to grow the fastest, since, as it were, they experience the highest ‘local’ *Rayleigh* number. As it happens, this wavelength, and the minimum Ra it requires to grow, come to:

$$\text{Ra}_{\text{crmin}} = \frac{27\pi^4}{4} \approx 657.5 \quad (3.18)$$

$$\lambda_{\text{cr}}^* = 2\sqrt{2} \approx 2.828 \quad (3.19)$$

At first glance it might seem that we have not truly answered the question of what defines the critical Rayleigh number for a convecting system as a whole, but rather only a contingent answer depending on wavelengths of perturbation. Consider, though, the significance of driving the Rayleigh number below the minimum critical value. This is equivalent to stating that no perturbations at all - not even the least stable ones - are able to grow quicker than the diffusive timescale. At the minimum critical value itself, it follows that only perturbations of $\sqrt{2}$ scale will grow; this value nonetheless serves adequately as the Ra_{cr} of the entire fluid, since a perturbation of such a wavelength can always be discovered in any real system - if geometry permits.

Having determined the conditions under which the conductive planform becomes unstable, it remains to establish the new stability criterion that system now seeks. Assuming that the fastest-growing perturbation will ultimately come to dominate all others, what we need is an expression for the velocity field in terms of λ that we can solve for the critical wavelength λ_{cr} [Rayleigh, 1916].

First let us find the infinitesimal thermal anomaly in terms of perturbation wavelength, which must be a sinusoidal function in both y and x :

$$\theta'^* = \tilde{\theta}_0^* \sin(\pi y^*) \sin\left(\frac{2\pi x^*}{\lambda^*}\right) \quad (3.20)$$

Where $\tilde{\theta}_0^*$ is the first term of the Fourier expansion of θ'^* and provides the wave amplitude, which is arbitrary.

We can now take the stream function Ψ in terms of θ' and substitute:

$$\Psi^* = -\left(\frac{\lambda^*}{2}\right) \left(\frac{4\pi^2}{\lambda^{*2}} + \pi^2\right) \tilde{\theta}_0^* \sin(\pi y^*) \cos\left(\frac{2\pi x^*}{\lambda^*}\right) \quad (3.21)$$

The contours of the stream function give the geometry of convection, which, for the critical λ in two dimensions, takes the form of pairs of counter-rotating half-cells of aspect $\frac{\lambda_{\text{cr}}^*}{2} = \sqrt{2}$; in other words, the planform of convection at steady state for any basally-heated planar isoviscous system will tend to approach an aspect ratio with the approximate dimensions, in landscape, of the page this sentence is written on.

Ra* in varying aspect ratios We have spoken thus far in terms of the critical *wavelength* of a perturbation. Whether that wavelength is actually *available* is another question. If we presume an infinitely wide plane domain populated with random fluctuations, it is self-evident that a perturbation of the appropriate wavelength will develop somewhere. Conversely, a domain whose aspect ratio is a finite value, A , is forced to make do with perturbations of a maximum wavelength of A . For the simple planar basally-heated isoviscous case, any value of A less than $\sqrt{2}$ would therefore be expected to suppress overall convective vigour by precluding the optimal perturbation wavelength. In effect we have introduced a dependency of Ra_{cr} on A such that [Malkus and Chandrasekhar, 1954]:

$$\text{Ra}_{\text{cr}} = \min_{A>0} \frac{\pi^4 (1 + A^2)^3}{A^4} \quad (3.22)$$

At any value of A greater than the ‘global’ critical wavelength of $\sqrt{2}$, we find that the critical *Rayleigh* number should come to:

$$\text{Ra}_{\text{cr}} = \frac{27\pi^4}{4} \approx 657.5 \quad (3.23)$$

By comparison, at the unit aspect ratios typically modelled ($A = 1$), the value of Ra_{cr} lies instead at [Grover and Tandon, 1968]:

$$\text{Ra}_{\text{cr}} = 2^3 \pi^4 \approx 779.3 \quad (3.24)$$

A value which is borne out in laboratory testing [Whitehead et al., 2011].

3.1.4 Boundary layer theory and the Ra – Nu scaling

While heat may be transported by convection in the interior of the system, heat may only cross in or out of the system as a whole via conduction. This occurs across two thin layers at the outer and inner boundaries. Since we stipulate that these layers are purely conductive, a *Rayleigh* number defined only across each layer must be below the critical value for that layer: $\text{Ra}_{\text{layer}} < \text{Ra}_{\text{layer}_{\text{cr}}}$ [Olson, 1987]. This is the first observation of boundary layer theory, whence can be deduced the following fundamental power law relationship between the *Rayleigh* and *Nusselt* numbers [Schubert et al., 2001]:

$$\text{Nu} \propto \text{Ra}^\beta, \quad \beta \rightarrow \frac{1}{3} \quad (3.25)$$

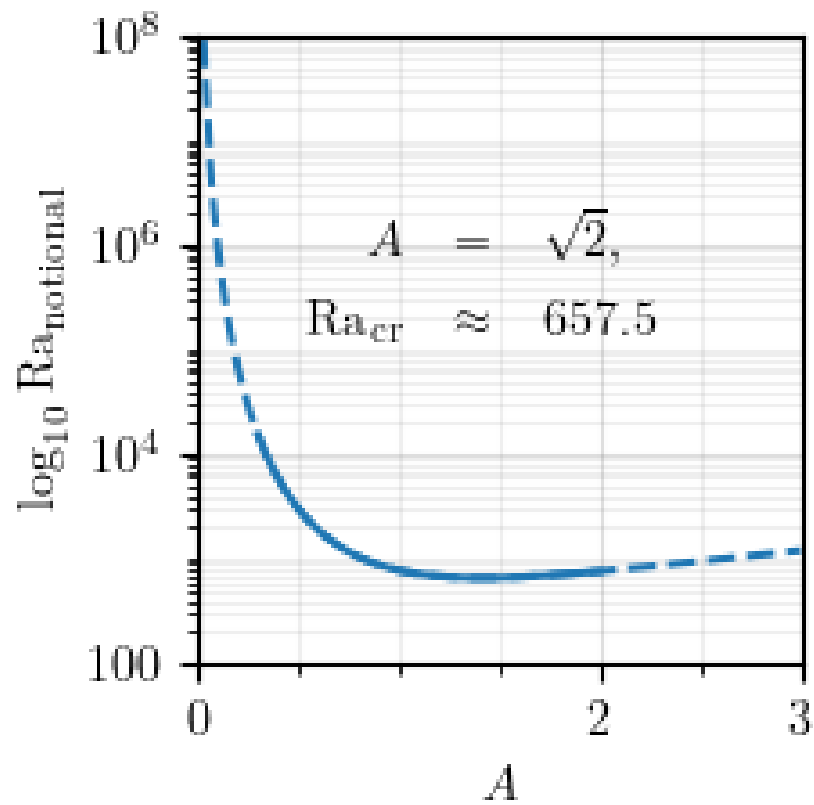


Figure 3.1: The global minimum of the curve of aspect ratio to ‘notional’ *Rayleigh* number gives the ‘critical’ *Rayleigh* number below which convection yields to conduction.

Where Nu is the Nusselt number. The coefficient of proportionality is theoretically ≈ 0.1941 [Olson, 1987], though it has been argued that its value will tend to be dominated by uncertainty in practice [Lenardic and Moresi, 2003]; reported values have ranged between $0.25 - 0.27$ [Olson, 1987, Jarvis and Mitrovica, 1989].

An equivalent scaling [Jarvis and Peltier, 1982] has instead:

$$Nu \propto Ra^{*\beta} \quad (3.26)$$

Where Ra^* , again, is the proportion by which Ra exceeds Ra_{cr} .

Defining Ra in this way preserves the value of β insofar as Ra_{cr} is independent of it, but allows the coefficient of proportionality to relate more strictly to non-thermal factors like the domain geometry - for example the aspect ratio, which (above a certain threshold) has been observed to stretch or compress the planform horizontally without changing the underlying boundary stability criteria [Jarvis and Peltier, 1982].

Another neat implication of the β law is that $Nu \rightarrow 1$ as $Ra^* \rightarrow 1$. This is intuitive because $Nu \rightarrow 1$ when convection collapses, which is just what is implied by $Ra^* \rightarrow 1$. However, we do have to be careful when considering subcritical values of Ra , which - if the β scaling is taken literally - would drive Nu to values below one: a sub-conductive geotherm. While a scenario like this could conceivably obtain over the short term - potentially even as a consequence of abrupt convective failure, as the scaling law suggests - it could not persist over the long term due to the relatively rapid rate of thermal diffusion relative to planetary timescales. Thus it is arguably inappropriate for a steady-state scaling law to project $Nu < 1$ in any scenario whatsoever. A branching function is implied where Nu tracks the conductive geotherm in subcritical conditions.

$$Nu = \begin{cases} Ra^{*\beta} & Ra^* > 1 \\ 1 & k \geq 1 \end{cases} \quad (3.27)$$

A conditional definition like this implies a piecewise function. If the relation is in fact continuous, as would be natural, the behaviour at the limit $Ra^* \rightarrow 1$ is unclear. In practice, as we have discussed, the fate of any convecting system near this boundary will be heavily controlled by the particular distribution of hot and cold material at any given time.

In the state where Nu satisfies the β law, the interior of each cell becomes a homogeneous region of uniform temperature T^{cell} and variable but low velocities, with strong gradients and shears at the margins, and overall cell dimensions approaching an aspect ratio of $\sqrt{2}$. Because of the fixed temperature scale, the only way heat transport can be enhanced in such a system is by thinning the boundary layers, which in practice occurs by dripping/pluming until only the theoretical stable boundary thickness is left. For this reason, Nu also functions as a useful proxy for boundary layer thickness when this is otherwise hard to define.

The canonical β scaling is seductive because it connects the relatively well-constrained fact of surface geothermal flux with the more mysterious thermal state of the mantle, and so allows parameterised thermal histories to be projected through deep time. The $\beta \rightarrow \frac{1}{3}$ limit itself ultimately derives from the *Rayleigh* number's dependence on length cubed, and while there is no *a priori* reason to believe that this analytical justification must be borne out in practice, it has been recognised as highly suggestive for over half a century [Chan, 1971]. Testing this scaling behaviour empirically was an early priority of computational geodynamics, with several studies producing estimates that converged on, but did not achieve, the theoretical $\frac{1}{3}$ scaling: the value has been reported as any of 0.313 [Jarvis and Peltier, 1982], 0.318 [Jarvis and Peltier, 1986], 0.319 [Schubert and Anderson, 1985], 0.326 [Jarvis and Mitrovica, 1989], 0.36 [Quarenì et al., 1985], and 0.31 [Niemela et al., 2000], using various methods both numerical and laboratory-based. The reason for the deviation is uncertain. One possibility is that the boundary layer instability theory is only valid in the limit $Ra \rightarrow \infty$ [Olson, 1987]. Alternatively, high Ra values may witness transitions to alternate scaling logics altogether - perhaps lowering β . It has even been suggested that, at very high *Rayleigh* numbers, an 'asymptotic regime' of $\beta \rightarrow \frac{1}{2}$ might emerge, though this has yet to be observed in practice [Niemela et al., 2000].

While the β scaling strictly holds only for those isoviscous systems with purely basal (no volumetric) heating, Cartesian geometry, and free-slip boundaries, it has been found to hold for a wide range of

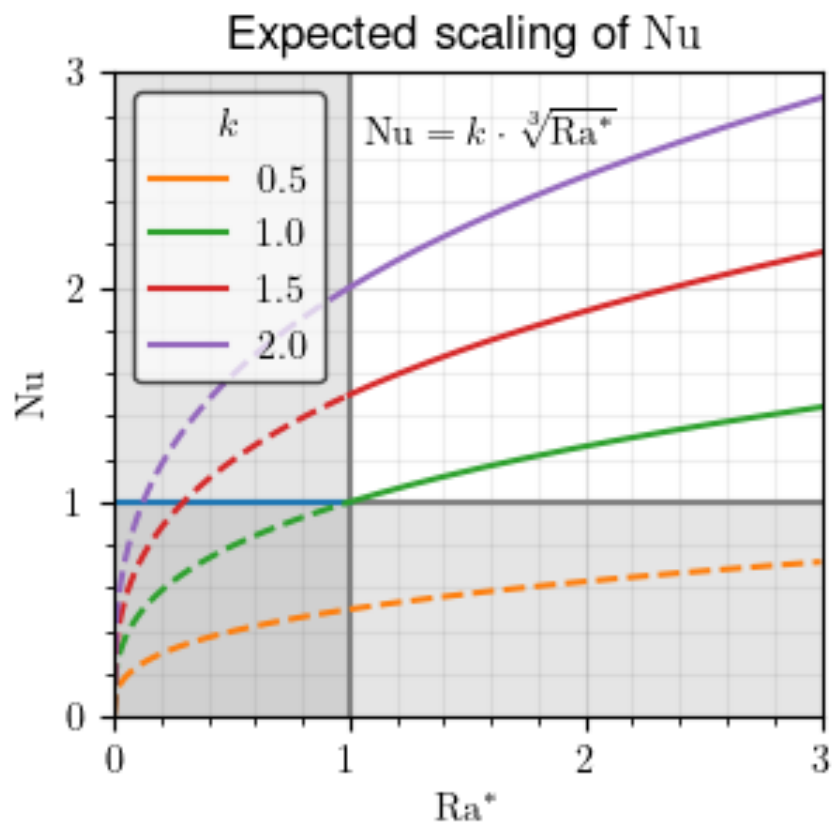


Figure 3.2: The β scaling for varying values of the proportionality constant k . When either R or Nu fall below one, a conductive geotherm is implied.

systems if certain corrections are made [Schubert et al., 2001].

In search of β : an analytical approach Much of this thesis will be dedicated to empirically obtaining values of β for systems of progressively increasing complexity. Once again, however, there is an analytical road to obtaining β for simple cases, which we owe with gratitude to the substantial textbook literature predating the computer age.

We have already established that when the *Rayleigh* number is supercritical, heat may be more rapidly transported by advection than by diffusion. The effect of the ensuing convection is to deflect this conductive geotherm T_c^* towards the steeper ‘adiabatic geotherm’: the path in temperature-pressure space along which the potential temperature θ - that which a parcel would achieve if brought to a standard reference pressure without gaining or losing any heat - is effectively constant. As the adiabat approaches the system boundaries, a point of diminishing returns is reached, and conductive processes take precedence once more. These regions of conductivity are the convecting system’s boundary layers.

It is possible to obtain an expression for the thickness of these boundary layers by considering the linear stability of just the layers themselves. First, we must determine the rate at which the conductive layer expands. This is complicated in the first instance by the fact that the actual layer thickness itself is hard to define in a continuum. Traditionally, however, it has sufficed to define it as the domain across which the first ten percent of temperature is gained or lost. Hence:

$$z_T = 2\eta_T\sqrt{\kappa t} \approx 2.32\sqrt{\kappa t} \quad (3.28)$$

Where z_T is the boundary layer thickness, $\sqrt{\kappa t}$ is interpreted as the characteristic length scale of thermal diffusion κ , and η_T is the inverse error function of 0.1, a constant term approximately equal to 1.16.

As the boundary grows, so do the thermal buoyancy forces. The relevant *Rayleigh* number to parameterise the vigour of the incipient convection is taken over the boundary layer thickness itself, and hence grows as the layer grows:

$$\text{Ra}_{z_T} = \frac{\alpha \Delta T g z_T^3}{\nu \kappa} \quad (3.29)$$

Where α is the thermal expansivity and ν is the dynamic viscosity $\frac{\nu}{\rho}$.

Now what we are interested in is what the thickness of the boundary layer will be when the *Rayleigh* number defined over it, Ra_{z_T} , is at its critical value, $\text{Ra}_{z_T \text{ cr}}$. Below this value, convective disruption of the layer will not be possible, as any perturbations within the layer will be thermally diffused before they can grow; while above this value, convection is inevitable and the conductive profile of the layer cannot be sustained. The expression for $\text{Ra}_{z_T \text{ cr}}$ is the same as that for Ra_{z_T} , except that the temperature contrast T is half that of the system as a whole; this is because the dimensionless temperature change across either boundary layer goes from zero or unit at the outer edge to exactly 0.5 at the inner edge, where the layers face the tepid conditions of the intracellular fluid; so we write:

$$\text{Ra}_{z_T \text{ cr}} = \frac{\text{Ra}_{z_T}}{2} \quad (3.30)$$

And accordingly:

$$z_T = \left\{ \frac{2\text{Ra}_F \nu \kappa}{\alpha g \Delta T} \right\}^{\frac{1}{3}} \quad (3.31)$$

Where Ra_F is coined to refer to the minimum critical *Rayleigh* number across the layer as defined when that layer is at the brink of collapse.

At this point we might be tempted to define a general critical *Rayleigh* number for the layer by the same means we deduced one for the system as a whole previously. Unfortunately, the dynamic quality of the layer thickness z_T poses one unknown too many. For a boundary layer that is developing through time, it is not guaranteed that an appropriate perturbation of the appropriate scale will emerge at the appropriate moment, nor even that the geometry of the layer will ever be sufficient to permit such a

perturbation in the first place. We have come as far as analytical methods can take us; to close the loop, it is necessary to obtain Ra_F empirically:

$$z_T = \left\{ \frac{807\nu\kappa}{\alpha g \Delta T} \right\}^{\frac{1}{3}} \quad (3.32)$$

Where the value 807 is the experimentally determined Ra_F for a free-slip surface [Jaupart and Parsons, 1985].

Now, because the thickness of a conductive layer is directly related to the thermal gradient across it, and thence to the Nusselt number Nu , while the right side contains the coefficients of the global Rayleigh number Ra , it finally becomes apparent what form the relationship between Nu and Ra should take:

$$Nu = 0.112Ra^{\frac{1}{3}}, \quad Ra_F = 807 \quad (3.33)$$

Or more generally:

$$Nu \propto Ra^{\beta}, \quad \beta \approx \frac{1}{3} \quad (3.34)$$

That a scaling law of this form would obtain for two dimensionless flow constants such as these is not surprising; empirically, just such a relationship is in fact very widely attested [Turcotte and Oxburgh, 1969, McKenzie et al., 1974, Solomatov, 1995]. Authors have differed, however, on the proper value of β . Though the canonicity of the analytically-derived value of one third is beyond dispute, it is clear from the divergent results of numerous studies that, in any real scenario, many more variables than we have accounted for must enter the equation. Time-dependence, long-lived thermal heterogeneities, aspect ratio, internal heating, and countless other factors all have a part to play. Obtaining robust scaling laws that account for all these factors is the vexing business of this thesis.

3.1.5 Chaos and attraction: approximate solutions to insoluble equations

The nature of convecting systems in practice ensures that even the simplest problems can be effectively or absolutely insoluble by analytical means. Though we will shortly outline methods for meeting these challenges experimentally, it is always unwise to go too far empirically whither mathematics cannot follow.

One means of probing beyond the insolubility barrier is to take an eigenmode expansion of the equations of state and discard all but the fewest number of terms which still support nonlinear interactions:

$$\begin{aligned} \Psi^* &= \frac{4 + \lambda^{*2}}{\sqrt{2}} A(\tau) \sin\left(\frac{2\pi x^*}{\lambda^*}\right) \sin(\lambda y^*) \\ \theta^* &= \frac{1}{\pi r} \left[C(\tau) \sin(2\pi y^*) - \sqrt{2} B(\tau) \cos\left(\frac{2\pi x^*}{\lambda^*}\right) \sin(\pi y^*) \right] \end{aligned}$$

Where Ψ is again the stream function, x and y are coordinates, λ is the featural wavelength, Ra^* is the *Rayleigh* number as a proportion of the critical value $Ra^* = \frac{Ra}{Ra_{cr}}$, and $A(\tau)$, $B(\tau)$, and $C(\tau)$ are time-dependent coefficients which are functions of τ , time non-dimensionalised by wavelength:

$$\tau = \pi^2 \left[1 + \frac{2}{\lambda^*} \sin(\pi y^*) \right] t^* \quad (3.35)$$

The A , B , and C coefficients permit a powerful simplification in form. Selecting the appropriate equation from the infinite set contained in the eigenmode expansion [Schubert et al., 2001], the following first-order differential equations can be obtained:

$$\begin{aligned}\frac{dA}{d\tau} &= \text{Pr} (B - A) \\ \frac{dB}{d\tau} &= rA - B - AC \\ \frac{dC}{d\tau} &= -bC + AB\end{aligned}$$

Where Pr is the *Prandtl* number, which must be kept finite for this analysis, though it may still be arbitrarily large; b represents:

$$b = \frac{4}{\left[1 + \left(\frac{2}{\lambda^*}\right)^2\right]} \quad (3.36)$$

These three are the Lorenz Equations [Lorenz, 1963], for which solutions represent states of cellular 2D convection. Because they are severely truncated in form, their scope of validity is limited to low values of Ra^* . Nevertheless, they are conceptually extremely useful for characterising the macro-scale character of mantle convection, particularly approaching the point of criticality.

Of the three functions, A relates to the stream function, B the resultant temperature variations, and C a horizontally averaged temperature mode. Three obvious solutions to the system are:

$$\begin{aligned}A = B = C &= 0 \\ A = B &= \pm \sqrt{b(r-1)} \\ C &= \text{Ra}^* - 1\end{aligned}$$

When $\text{Ra}^* < 1$, the trivial first solution above describes the only stable steady-state solution and represents pure conduction, just as we would expect when $\text{Ra} < \text{Ra}_{\text{cr}}$. When $\text{Ra}^* > 1$, this solution becomes unstable, and the only stable solutions become the positive and negative valencies of the second expression above, which represent clockwise and counterclockwise unicellular convection. The ‘choice’ of the system to devolve from the unbiased conductive solution to one of either the left- or right-biased convective solutions is termed a ‘pitchfork bifurcation’, the first of many we will encounter; its existence proves that mantle convection is chaotic.

The two convective solutions above have been shown to be stable - but are they necessarily steady? If we take our two primitive convective solutions further, a characteristic equation can be obtained from which we can derive the following special value of Ra^* :

$$\text{Ra}^* = \frac{\text{Pr} (\text{Pr} + b + 3)}{\text{Pr} - b - 1} \quad (3.37)$$

When $\text{Pr} > b + 1$, the above expression gives the value of Ra^* above which the two fundamental convective solutions are in fact not stable; in other words, it is the criterion for the instability of steady convection. It is also another kind of bifurcation - a *Hopf* bifurcation. Around a *Hopf* point, stable solutions are periodic and cyclical; solutions which cross the bifurcation are hence ‘captured’ by it and cycle through a finite set of states ad infinitum, until or unless those oscillations become great enough to tip a system into the zone of attraction of another *Hopf* point. Complex paths through phase space can thus be drawn which represent very high-order periodic solutions for the system that resist analytical description. Dubbed ‘strange attractors’, they are the iconic property of chaos theory.

So far, for the sake of argument, we have assumed a finite *Prandtl* number. This of course contravenes one of the foundational assumptions of our broader analysis. Before moving on, it behooves us to ask whether the chaotic behaviours observed in the Lorenz equations hold in the limit that $\text{Pr} \rightarrow \infty$.

This would imply, first of all, that $A = B$. Hence:

$$\begin{aligned}\frac{dA}{d\tau} &= \frac{dB}{d\tau} = (r-1)B - BC \\ \frac{dC}{d\tau} &= -bC + B^2\end{aligned}$$

The fixed points of these new equations are the same as for the Lorenz equations, as is the conductive solution when $A = B = C = 0$, which as before is stable only for subcritical Ra ; however, the convective solutions can be shown to be stable for all $r > 1$. We might take this to imply that mantle convection cannot be chaotic after all. However, it must be recalled that the Lorenz analysis begins with severe truncation of non-linear terms. For higher-order truncations, it is evident that chaotic phases can exist [Schubert et al., 2001], particularly at high *Rayleigh* numbers; what is not certain is whether, for a given degree of truncation and a given range of parameters, chaotic behaviours will manifest for a particular system. When we attempt to engage with the problem numerically and empirically through modelling, which is the purpose of this thesis, it will be seen that certain parameter bands are chaotic and time-dependent while others are not; ultimately it will be argued that such zones of chaos represent boundaries in a very high-dimensional phase space, and relate fundamentally to the nature and proper characterisation of tectonic modes.

3.2 Incorporating radiogenic heat: the internal- and mixed-heating cases

On Earth and other planets, the geothermal flux is not solely driven by basal heating from the core: there is a substantial, variable, and transient contribution from radioactive decay as well [Daly, 1980]. In mantle convection studies this is dubbed ‘internal heating’, and a system driven by both basal and internal heating is said to be ‘mixed-heating’.

The ratio of internal heat production to surface heat flux inside a planet is called the *Urey* ratio Ur [Urey, 1955]. Values less than one signify that a planet on the whole is cooling, while values greater than one represent global interior warming. To avoid projecting unreasonable global temperatures into Earth’s past, it was originally calculated that, for a *beta* exponent of $1/3$, the *Urey* number must be greater than ~ 0.7 [Christensen, 1985]. This assumption was forced into revision when it transpired that the radiogenic inventory of bulk mantle materials is far too low to support such a high radiogenic fraction [Jochum et al., 1983]. Direct evidence from geoneutrinos now suggests that the *Urey* ratio must be around 0.5 [Gando et al., 2011], and further argues that heat production is non-uniform throughout the mantle [Huang et al., 2013b]. Thermal histories that honour these facts have proved very difficult to construct [Korenaga, 2003, 2008, Mareschal et al., 2012, Dye, 2012, Jaupart et al., 2015, Korenaga, 2017].

Analytically, the addition of internal heating introduces many complications. Most significantly, the temperature scale is no longer an independent variable, but a natural one emerging from the dynamic balance of heat production and heat transport. This dependency invalidates the derivation of Ra previously provided, and unfortunately, no obvious or unambiguous alternative scaling is known [Schubert et al., 2001, Moore, 2008, Korenaga, 2017].

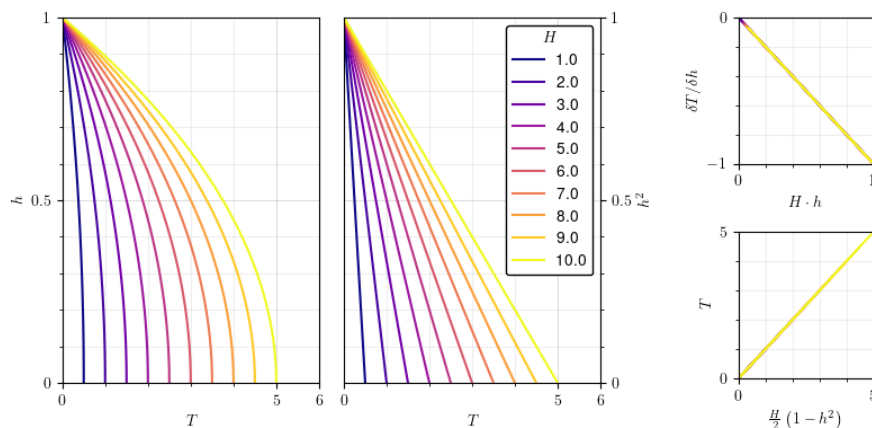


Figure 3.3: Summary of the scaling behaviours of isoviscous conduction for varying internal heating parameter H . Temperature goes linearly as a function of dimensionless height from the base of the mantle, h^2 .

3.2.1 Internal heating

To provide some bounds on the expected behaviour of a mixed-heating model, it is helpful to review the effects of purely internal heating. Without the ability to arbitrarily specify a temperature scale, it is natural to use that which arises from pure conduction. Under a scenario of pure internal heating, the conductive geotherm is no longer linear, but scales with length squared [Turcotte and Schubert, 2014] Figure 3.3:

$$\begin{aligned}\frac{dT_c}{dh} &= H \cdot h \\ \therefore T_c(h) &= \frac{H}{2} (1 - h^2)\end{aligned}$$

Where H is the per-mass heating rate and h is the dimensionless height from the base of the mantle. The maximum mantle temperature possible for a given value of H is thus $H/2$. Because, at the outer boundary, the thermal flux is exactly H (or, technically, $H \cdot \rho$ if dimensionalised), H here is identical to the ‘conductive *Nusselt* number’, Nu_c .

Having obtained the conductive geotherm, we can select a nominal temperature scale:

$$\Delta T_H = \frac{\rho H b^2}{k} \quad (3.38)$$

Where k is the conductivity, ρ is the density, H is the per-mass heating rate, and b is the characteristic spatial length. The temperature scale allows an ‘internally heated *Rayleigh* number’ Ra_H to be defined [Roberts, 1967]:

$$\begin{aligned}Ra_H &= Ra_B \cdot \Delta T_H \\ &= \frac{\alpha g \rho H b^5}{k \kappa \nu}\end{aligned}$$

Where Ra_B is the *Rayleigh* number derived for basal heating, introducing the parameters of thermal expansivity α , gravity g , thermal diffusivity κ , and momentum diffusivity ν [Turcotte and Schubert, 2014]. As in the basally-heated case, there exists for this number a critical value Ra_{Hcr} which divides the purely conductive regime from the convecting regime [Roberts, 1967, Schubert et al., 2001]:

$$Ra_{Hcrmin} = 867.8, \quad \lambda^*_{cr} = 3.51 \quad (3.39)$$

In other words, for the onset of convection in an internally-heated system with basal-insulating, surface-isothermal, free-slip boundaries, the critical *Rayleigh* number and characteristic wavelength are both a little more than one quarter greater than for the equivalent basally-heated case.

Usually we would be interested in deriving some scaling for the *Nusselt* number with respect to Ra_H . In this case, however, Nu must always equal one. This follows because there are no other sources or sinks in the model except for H , which must be radiated in full from the upper boundary whether the interior is convecting or not. Rather than serving to augment the surface flux, convection in a purely internally-heated system only functions to smooth out interior temperatures. To describe this, we must choose a representative internal temperature T_v , which should reflect a typical temperature in those regions where advection dominates [Solomatov and Moresi, 2000]. Normalising T_v by the true (measured) temperature scale ΔT gives T_v^* , the non-dimensional internal temperature, which then informs the definition of an ‘internal *Rayleigh* number’ Ra_v :

$$Ra_v = \frac{\alpha g T_v^* \Delta T b^3}{\kappa \nu} \quad (3.40)$$

This observes, in other words, that - in an internally-heated system - the thermal contrast available to drive convection will always be less by some factor than the thermal contrast across the system as a whole.

While the flux across the upper boundary is fixed by the choice of parameter H , the actual geometry of the boundary that supplies this flux must be a function of the *Rayleigh* number. At steady state, the interior temperature and the layer thickness adjust until a balance is reached, resulting in an outer temperature drop ΔT_o and outer boundary thickness δ_o that scale with, but are not solely determined by, the heating rate [Schubert et al., 2001]:

$$\Delta T_o \propto \frac{H^{3/4}}{\text{Ra}^{1/4}}$$

$$\delta_o \propto \frac{1}{(\text{Ra}_H)^{1/4}}$$

Wherein we observe that the implied flux (temperature drop divided by boundary thickness) remains equal to $\text{Nu}_c = H$, and hence $\text{Nu} = 1$, as expected: all of that inner motion occurs without leaving any thermal trace on the surface.

By taking a boundary layer approach to the heat-producing regions [Jaupart and Mareschal, 2010] and exploiting certain known requisites of the purely conductive state [Vilella and Kaminski, 2017], one can derive an alternative statement of the expected outer boundary layer properties in terms of the critical *Rayleigh* number:

$$\frac{\delta_o}{h} = \left(\frac{\text{Ra}_{H_{\text{cr}}}}{\text{Ra}_H} \right)^{1/4}$$

$$\frac{\Delta T_o}{\Delta T_H} = \frac{1}{2} \frac{\delta_o}{h}$$

This relationship appears to hold empirically with good confidence [Vilella and Kaminski, 2017].

Qualitatively, the addition of internal heating to the isoviscous convection problem imposes an asymmetry between upwellings and downwellings, which would otherwise be temperature-reversed mirrors of one another [Weinstein and Olson, 1990]. Because the thermal gradient of the conductive state rapidly drops with depth, the local *Rayleigh* numbers of deeper layers can quickly become subcritical, such that a large portion of the domain from the base up is locked in a near-conductive state. At high H , large-scale motions cease, and only a thin sub-surface ‘mixing layer’ witnesses any meaningful convection at all [Parmentier et al., 1994].

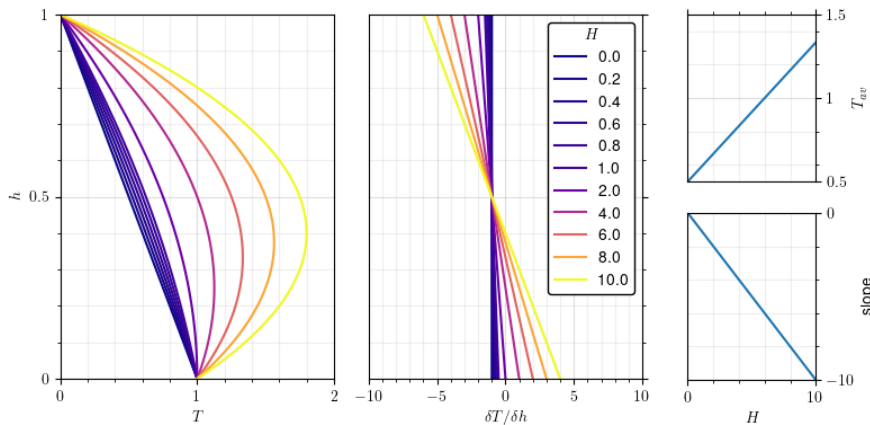


Figure 3.4: Summary of the scaling behaviours of the conductive solution for mixed heating. The inner boundary is set to dimensionless temperature $T = 1$, the outer to $T = 0$. On the right, average temperature and the slope of $\delta T / \delta h$ are given as functions of H . Gradient clearly varies as a function of dimensionless height h above the mantle base according to a slope given by $-H$. The conductive geotherm for mixed heating is therefore, in fact, a parabola.

One final feature of purely internal heating systems worth mentioning - obvious when pointed out but easily overlooked - is that the total heat production is fixed. This is *not* the case for basal

heating (using a fixed lower boundary temperature) because one can always, as it were, ‘suck’ more heat out of the core by thinning the lower boundary layer. Consequently, for a given ‘total heating rate’ $H_{\text{total}} = H \cdot \text{Area}$, the whole domain must necessarily be warming, cooling, or at equilibrium depending on the ratio of H_{total} and ϕ_{q_c} . This is just another reason to add to the many already presented for why we should expect purely internally-heated models to be less dynamic in every way than basally-heated ones.

3.2.2 Mixed heating

For systems driven by both internal and basal heating, the choice between basally-derived and internally-derived *Rayleigh* numbers leads to ambiguity and confusion. One path forward is to define a dimensionless heating rate H [Schubert et al., 2001]:

$$H = \text{Ra}_H / \text{Ra} \\ = \frac{\rho H^* D^2}{k \Delta T^*}$$

The conventional basally-heated Ra derivation can then be used, with H as a correcting coefficient.

While the purely internally-heated case has only one boundary layer (the outer), and the purely basally-heated case has symmetrical outer and inner boundaries, the mixed-heating case has two asymmetrical boundaries which are at least pseudo-independent of each other. While, at steady-state, the outer boundary flux is constrained to be always greater than the inner, the inner boundary flux may freely adjust itself as a function of the temperature differential between the prescribed lower boundary condition and the interior temperature. This intrinsic feedback makes quantitative analysis much more complicated.

As usual, a close consideration of the situation at the boundaries can set us on the right course. In the sorts of scenarios relevant to planetary settings, the lower boundary will remain hotter than the interior and the flux will be positive and substantial, in which case the outer boundary is required to transport the volumetric heating H in addition to the basal heating [Moore, 2008]:

$$\phi_{q_o} \propto \left| \frac{dT}{dh} \right|_{h=1} \\ \propto H + \phi_{q_i} \propto H + \left| \frac{dT}{dh} \right|_{h=0}$$

Where ϕ_{q_o} and ϕ_{q_i} are the inner and outer heat fluxes.

At the limit where internal heating becomes negligible compared to the inner-outer temperature drop, internal heating can be ignored altogether and the system approximates a purely basally-heated model. At the opposite extreme, if the interior temperature due to internal heating is high enough, the inner flux may become negative, and the mantle will cool into the core: a case hard to envision in nature. Between these regimes is a transition point when the interior temperature becomes equal to the lower boundary condition. In this borderline case, the inner (core-mantle) flux is exactly zero (i.e. insulating), so that the mixed-heating system reproduces a purely internally-heated system. Thus, the mixed-heating model includes both internally-heated and basally-heated endmembers.

To sharpen this analysis, we can look at more closely at the conductive profile. While it is possible to derive this from first principles, it is more easily illustrated by a numerical approach Figure 3.4. It is clear that the conductive geotherm forms a parabola whose second derivative is exactly equal to $-H$; the rest follows by integration, with the constants provided by logic and observation:

$$\begin{aligned}
T_c(h)'' &= -H \\
T_c(h)' &= -H \left(h - \frac{1}{2} \right) - 1 \\
T_c(h) &= \frac{Hh}{2} (1 - h) - h + 1 \\
T_{cav} &= \frac{H}{12} + \frac{1}{2}
\end{aligned}$$

If each geotherm traces a parabola, a maximum ‘natural’ temperature is implied where the first derivative is zero:

$$h_{T_{\max}} = \frac{1}{2} - \frac{1}{H}, \quad H > 0 \quad (3.41)$$

If $h_{T_{\max}}$ is less than zero for a particular value of H , then that maximum will never be realised and the true maximum temperature will be that at the mantle base, which is unit in our dimensionless treatment. The condition $h_{T_{\max}} > 0$ thus represents the regime boundary between those conductive solutions that cool into the core and those that only cool into space. This occurs at exactly $H = 2$, or more generally:

$$H_{\text{cr}} = \frac{1}{T_{cav}(H=0)}, \quad \text{Ra} < \text{Ra}_{\text{cr}} \quad (3.42)$$

Where H_{crit} is the critical heating factor above which the mantle cools into the core and at which the lower boundary becomes effectively insulating, as previously discussed.

If we define a ‘conductive *Nusselt* number’ - something of an oxymoron - we can see how the true *Nusselt* number should be expected to scale:

$$\begin{aligned}
\text{Nu}_{c(\text{mixed})} &= -T_c(h)', \quad h = 0 \\
&= 1 + \frac{H}{2} \\
&= \text{Nu}_{c(\text{basal})} + \text{Nu}_{c(\text{internal})}
\end{aligned}$$

I.e. the surface flux in the mixed case is simply the sum of the two heat drivers considered separately, just as we would expect from first principles.

We now have an understanding of how the system behaves below the critical *Rayleigh* number; now we must go above. One thing we can immediately observe from first principles is that the outer heat flux should be equal to the balance of H and whatever flux, positive or negative, is occurring over the lower boundary:

$$\phi_{q_o} = H + \phi_{q_i} \quad (3.43)$$

To analyse the mixed heating case, then, we need to invert our perspective and focus not on the outer boundary but on the inner boundary, whose core-ward temperature is equal to 1 and whose mantle-ward temperature - the ‘interior temperature’ T_i^* - is to be determined. Unfortunately, no universally accepted analytical treatment of interior* post-convection has yet been devised.

One workaround starts with the interior temperature analytically obtained for a purely internally-heated system and modify it with empirical constants to accommodate the addition of basal heating [Moore, 2008]:

$$T_{\text{interior}} = 0.49 + 1.24H^{3/4}\text{Ra}^{-1/4}, \quad H < H_{\text{inv}} \quad (3.44)$$

This scaling, however, is back-formed from an observed and arbitrary measurement of T_{interior} taken across the middle 60% of a finite-element numerical model, and no justification for this particular choice is ventured nor defended across the range of Ra and H values sampled.

Given a known value of T_{interior} , the associated *Nusselt* number should be proportional to that temperature drop divided by the stable outer boundary thickness δ_o . Conventional boundary layer analysis can be used to argue a scaling of the form [Schubert et al., 2001]:

$$\delta_o = \left(\frac{\text{Ra}}{\text{Ra}_{\text{cr}}} \frac{\Delta T_b}{\Delta T} \right)^{-1/3} \quad (3.45)$$

Empirically, the same authors [Moore, 2008] found that this scaling holds only for a much lower exponent of ~ -0.303 for $\text{Ra} \leq 10^8$ and $H < 10$, while apparently approaching the theoretical value with increasing H ; the discrepancy was attributed to disruption by plumes from the lower boundary.

The next step is to attempt to derive the *Nusselt-Rayleigh* scaling, which we expect to approach the conventional power law behaviour under the *beta* exponent $\beta = 1/3$. The expected relationship, however, does not manifest even approximately for the mixed heating case unless alternative definitions of Nu and Ra are used. Subtracting the basal component from Nu and using the arithmetic difference between Ra and Ra_{cr} yields the following with some confidence [Moore, 2008]:

$$\text{Nu} - 1 = \frac{H}{2} + 0.206 \cdot (\text{Ra} - \text{Ra}_{\text{cr}})^{0.318} \quad (3.46)$$

While the above was calculated using a planar 2D model, an equivalent model suite in the spherical shell closely reproduced this relationship after geometry was accounted for [Weller et al., 2016].

An alternative parameterisation [Vilella and Deschamps, 2018] expresses Nu in terms of the critical heating rate H_{crit} after convection:

$$H_{\text{cr}} = 2 + 2C_N \left(\frac{\text{Ra}}{\text{Ra}_{\text{cr}}} - 1 \right)^{1/3} \quad (3.47)$$

Where C_N is an empirically-obtained constant argued to be exactly equal to 1.5. This definition of the convective H_{cr} further defines two independent *Nusselt* numbers depending on the magnitude of heating:

$$\begin{aligned} \text{Nu} &= \frac{1}{2} (H + H_{\text{cr}}), & H &\leq H_{\text{cr}} \\ \text{Nu} &= H - \left(2 \frac{\Delta T_{\text{TBL},b}}{\Delta T} H \right)^{1/2}, & H &\geq H_{\text{cr}} \end{aligned}$$

Where $\Delta T_{\text{TBL},b}$ is the temperature jump across the inner boundary layer, recalling that the mantle must cool into the core above H_{cr} .

3.3 Thinking outside the box: building a cylindrical domain

Thus far we have restricted this discussion to rectilinear (‘Cartesian’) planar boxes. Real planets are of course three-dimensional balls, not two-dimensional boxes. While we are bound to the planar realm by the dictates of pragmatism, we can at least step towards realism by embracing a curved geometry. Indeed, it transpires that even this small step introduces substantial complications - and raises new and fascinating questions.

3.3.1 Establishing a coordinate system

In any convection model, gravity defines the natural ‘down’ direction and gives us our first most important scale: the depth z from the surface, or its complement, the height from the model base $h = 1 - z$.

If the domain is allowed to curve around a certain locus, a cylindrical or annular geometry is obtained which is more appropriate for planetary mantles. While we retain h and z as terms relevant to any action within the domain, we must also introduce a concept of radial height r , understood here to represent the distance from the planetary centre of gravity. The cylindrical domain, for us

representing the mantle, is thus bounded by the inner radius r_i and the outer radius r_o , defining an area of $\pi(r_o^2 - r_i^2)$.

Our choice of radii implies a degree of curvature f :

$$f \equiv \frac{r_o}{r_i} \quad (3.48)$$

Where $f \rightarrow 1$ is equivalent to an infinitely wide Cartesian box, $f \rightarrow 0$ represents a complete disc (i.e. no hole in the middle), and the values ~ 0.5 and ~ 0.9 would be appropriate for the whole mantle and upper mantle respectively. The ratio of radii f is identical to the ratio of circumferences, so that $f = 0.5$ represents a system where the arc length of the base is half that of the surface. (Note that this would imply infinite planetary radius at $f = 1$ - hence the planar-like endmember $f = 1$ is not strictly reachable under an assumption of curvature, though arbitrarily high values can be set to reproduce that behaviour [Jarvis, 1993])

If we further stipulate that the radial thickness of the domain is restricted to unit:

$$\Delta r = r_o - r_i = 1 \quad (3.49)$$

$$r_o \rightarrow 1 \quad \text{as} \quad f \rightarrow 0 \quad (3.50)$$

Then:

$$r_i = \frac{f}{1-f}, \quad r_o = \frac{1}{1-f} \quad (3.51)$$

$$r(h) = r_i + h \quad (3.52)$$

In short: honouring this constraint allows us to produce a workable radial coordinate system simply by setting a desired value of f .

At other times, it may be convenient to set the radius at the outer boundary as unit, and relax the constraint for the mantle thickness to be unit. This has the effect of scaling the inner radius so that it is exactly f . We will call this metric the ‘planetary radial scale’ r^* :

$$r_i^* = f, \quad r_o^* = 1 \quad (3.53)$$

$$\Delta r^* = 1 - f \quad (3.54)$$

$$r^*(h) = \frac{h + r_i}{r_o} = \frac{\Delta r^*}{r} = h(1 - f) + f \quad (3.55)$$

(Note that r^* and r converge as f approaches zero.)

This leaves us with four different terms to describe radial position: h , the dimensionless height from the mantle base; z , its complement; r , the radial scale such that the thickness of the mantle is one; and r^* , the radial scale such that the total planetary radius is one. Each of these scales will prove natural in some contexts and less so in others, and all find use in our analysis.

We have our radial coordinate system: now we need a system for our angular position too. The obvious way to do this is by simply providing an angle θ in radians anticlockwise from an arbitrary origin - i.e. $0 \leq \theta < 2\pi$. In practice, we will often want to work with only a small wedge of the planet at any given time. This is equivalent to choosing a maximum value, Θ :

$$0 \leq \theta < \Theta \leq 2\pi \quad (3.56)$$

If the simulation is to be interpreted as (implicitly) a piece of a global, radially symmetrical planform, values of Θ must fall within π/l , where l is any positive integer. This allows the domain to be mirrored and multiplied to cover the full disc without distortion (Figure 3.5).

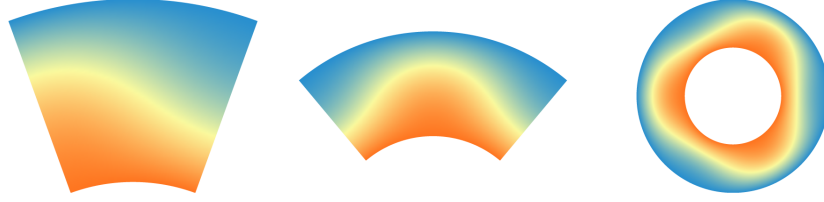


Figure 3.5: Illustration of the relationship between a wedge of an annulus and the full disc. We can tile the wedge across the whole disc by first mirroring it, then copying it. If we wish to avoid stretching or squeezing the original state to make it fit, we must ensure that Θ (angular extent of the wedge in radians) is a positive integer ratio of π . In this case, Θ goes from $\pi/3$ (left) to $2\pi/6$ (centre) to 2π (right: the full annulus).

In the same way that we built an artificial scale r for the purpose of normalising the radial thickness, we can also build a scale l for the width. This also gives us a chance to reverse the convention from anticlockwise (right-to-left) to clockwise (left-to-right), which is more familiar for Cartesian domains.

$$l = \frac{\Theta - \theta}{\Theta} \quad (3.57)$$

Defined this way, the coordinate pair (l, h) reproduces in the annulus the (x, y) coordinate system of a Cartesian unit square. This gives us a universal coordinate system for all cylindrical domains, regardless of curvature: allowing, for example, the ‘splaying’ of a Cartesian box model into an annular wedge, or the ‘squaring up’ of a wedge into a box.

When dealing with a Cartesian box geometry, one characteristic measure is the aspect ratio A , where for instance $A = 1$ would denote a square box and $A = 3$ a wide rectangle. If we wish to carry this measure into the cylindrical domain we need to choose a particular ring - a curve of constant depth - to be the characteristic angular length scale. The two most obvious candidates would be the outer and inner boundaries. However, it proves most convenient to take a different approach and instead draw an arc through the mid-depth, halfway (radially) between the outer and inner boundaries. The aspect ratio can then be defined as the length of this arc divided by the radial length. The mid-radius can be calculated from f :

$$r_m \equiv \frac{r_i + r_o}{2} = \frac{1 + f}{2(1 - f)} \quad (3.58)$$

Since the circumference of a complete circle is $2\pi r$ (we would hope no citation is needed for that one), the angular length at depth r_m can be calculated from Θ :

$$A = r_m \Theta \quad (3.59)$$

Such a scheme leaves us with two competing claims for a ‘natural’ denominator of the angular coordinate - Θ and r_m . While authors have sometimes preferred to keep Θ and r_m constant and allow A to vary [Jarvis, 1994], we have for the most part chosen to fix A and r_m with Θ as the free parameter, as in [Jarvis, 1993]. One of the virtues of this choice is that it preserves the (g, h) coordinate system over varying A . This simplifies comparisons with plane-layer simulations, though potentially at the cost of producing planforms which could be unstable if scaled to the full annulus.

In the Cartesian case, when the height of the box is set to unit, the aspect ratio is not only equivalent to the box width: it is also equivalent to the box *area*. The virtue of defining cylindrical A using the mid-depth is that this property is preserved even for extreme values of f . Parameterising a model in terms of area is particularly advantageous when dealing with system forcings, like internal heat, which scale with area.

While it is trivial to divide the domain in an angular sense (i.e. splitting the wedge into more wedges), dividing it in a radial sense requires a little more consideration. The proportion of the annulus lying below a particular depth - which we shall call *Disc* - is a function of the inner and mid-radii:

$$\text{Disc}(r) = \frac{r^2 - r_i^2}{2r_m} = \frac{r^* - f^2}{1 - f^2} \quad (3.60)$$

As we have already established that the total area will always equal the aspect ratio A , the true area under any depth can then be given simply as $\text{Disc} \cdot A$.

Laying the datum for the aspect ratio through the mid-depth also has the benefit of providing a good reference scale for the angular length, which allows us to set aside θ and Θ altogether and deal with both radial and angular distances in like units. Let s be the angular length at any given depth. We already know that $s_m = A$ by definition, but we can just as easily calculate s for any value of r :

$$s(r) = r\Theta = r \frac{A}{r_m} = rA \frac{2(1-f)}{1+f} \quad (3.61)$$

At low values of f (therefore high curvature), s is strongly dependent on r , with the inner surface much shorter than the outer surface. Conversely, at values of f approaching 1, the dependence on r disappears as the value of r_o becomes indistinguishable from r_i - in which case $s \approx A$ throughout the domain, as it does in a Cartesian box.

It will shortly prove convenient to non-dimensionalise s as $s^* = s/A$, such that the dimensionless length through the mid-depth $s^* = 1$. We can then write s^* very simply as a function of r^* and the inner and outer lengths accordingly:

$$s^*(r^*) = 2 \frac{r^*}{1+f} \quad (3.62)$$

$$s^*_i = 2 \frac{f}{1+f}, \quad s^*_o = 2 \frac{1}{1+f} \quad (3.63)$$

The length s is, among other things, the factor by which an average measurement of some variable taken across a layer can be converted into a total value for that layer. It is vital to account for varying s whenever comparing between different layers in a given system, or between equivalent layers in systems of differing f .

3.3.2 Conduction in the basally-heated cylindrical case

It is a requirement of thermal equilibrium that the thermal flux must be the same through every layer. In the planar case this results in a linear geotherm which, in a model with fixed and unitless boundary temperatures, results in a simple function of $T = z$ where z is dimensionless depth from the top of the model. The average temperature is then trivially $T_{\text{av}} = 0.5$. (For any system in pure conduction the *Nusselt* number is by definition 1.)

In a cylindrical domain, however, the length of each layer s is a function of depth and curvature as we have shown; consequently, shallower layers are able to transmit the same flux with a smaller temperature drop:

$$\phi_q = -s \cdot \frac{dT}{dh} \quad (3.64)$$

To define the flux, we need the geothermal gradient. The conductive geotherm can be elegantly stated in terms of r^* Figure 3.6 Figure 3.7:

$$T_c(h) = \frac{\ln r^*}{\ln f} \quad (3.65)$$

And so the geothermal gradient:

$$\frac{dT_c}{dh} = \frac{f-1}{r^* \ln f} \quad (3.66)$$

And finally the flux itself can be written as:

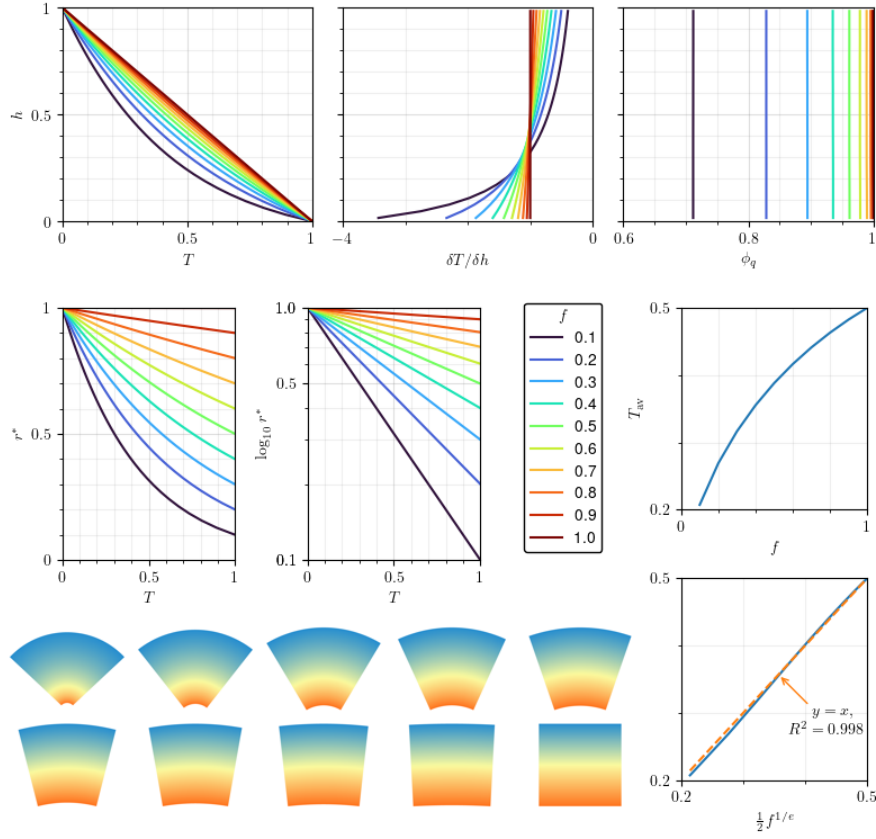


Figure 3.6: Summary of the scaling behaviours of isoviscous conduction for varying curvature parameter f . We obtain a natural scaling for f versus T_{av} with an R^2 better than 99%.

$$\begin{aligned}\phi_{qc} &= -\frac{s^*(f-1)}{r^* \ln f} \\ &= -\frac{2(1-f)}{(f+1) \ln f}\end{aligned}$$

$$\begin{aligned}&\rightarrow 1 && \text{as } f \rightarrow 1 \\ &\rightarrow 0 && \text{as } f \rightarrow 0\end{aligned}$$

Or very succinctly in terms of the ‘true’ radius of the mid-depth:

$$\phi_{qc} = -\frac{1}{r_m \ln f} \quad (3.67)$$

To facilitate comparison between systems of different curvature, we can then use the above to define a dimensionless planetary flux ϕ_q^* - which is really just another name for the *Nusselt* number Nu:

$$\begin{aligned}\phi_q^* &= \frac{\phi_q}{\phi_{qc}} \\ &\equiv \text{Nu}\end{aligned}$$

Where the subscript c , here as elsewhere, denotes a purely conductive endmember. Because Nu now inherits a dependency on f , it is no longer equivalent to the dimensionless surface temperature gradient, and so it is important always to present and discuss it in its proper terms as a ratio of fluxes.

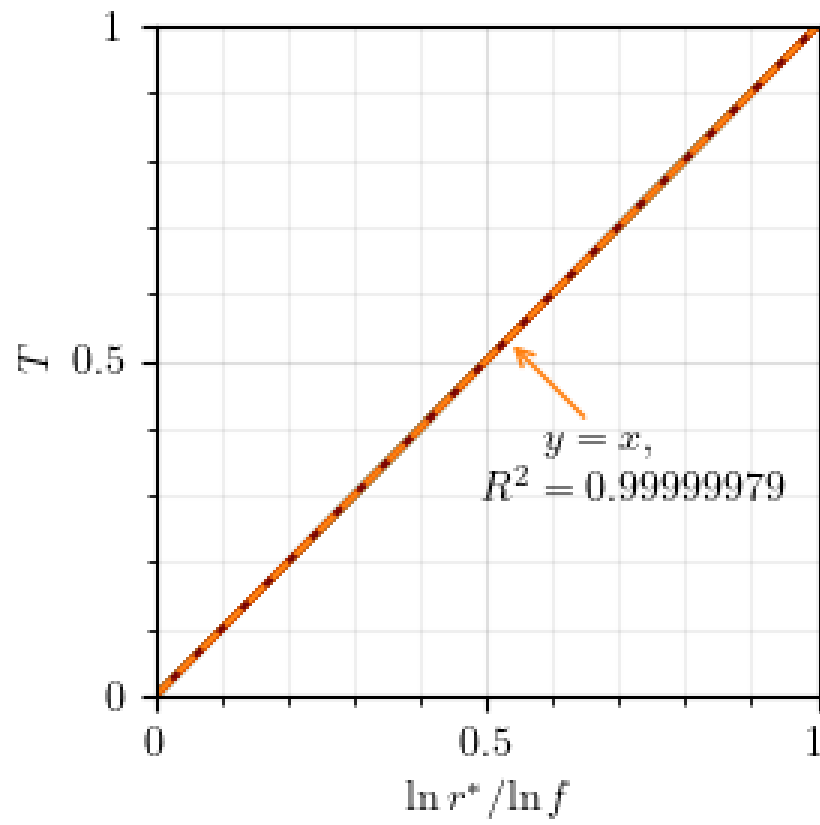


Figure 3.7: The analytical scaling of conductive temperature with $\ln r^* / \ln f$ holds empirically with extreme precision.

Just as the flux now scales with f , so must the average mantle temperature. In the planar case, the average temperature of the system is always half the temperature drop. In the cylindrical case, however:

$$\begin{aligned} T_{\text{av}} &= \frac{1}{2} \sqrt[2]{f} \\ &\equiv T_c \end{aligned}$$

The relationship is apparent in the numerical results Figure 3.6.

Instability and convection An implication of Nu 's dependency on curvature is that the upper and lower boundaries must no longer be symmetrical. This invalidates many of the assumptions that made the planar case amenable to analysis. The additional space at the top of the model now allows more room for downwellings relative to basal upwellings, tending to promote instability [Jarvis, 1991]; on the other hand, the curved geotherm and the increased surface for radiating heat would tend to permit a comparatively thicker upper boundary layer. The effect of these countervailing forcings on the fundamental scalings of Nu , Ra , Ra_{cr} , and the all-important relation $\text{Nu} \propto R^\beta$ is not obvious.

To begin to unpack the complexities of convection in the annulus, we can start with the assumption that - as in the planar case - the convective steady state will eventually result in a broad intracellular region of uniform temperature T_{cell} . Assuming a unit temperature drop $\Delta T = 1$, we can write:

$$\begin{aligned} \Delta T_o &= T_{\text{cell}} \\ \Delta T_i &= 1 - T_{\text{cell}} \end{aligned}$$

Knowing that the inner and outer fluxes ϕ_{q_i} and ϕ_{q_o} must be equal at steady state, and that the outer boundary - due to its greater length - can sustain that flux with a gradient shallower by a factor of f , we can deduce a relation between the outer and inner thermal gradients, and thence between T_{cell} and the inner and outer boundary layer thicknesses Δr_i and Δr_o :

$$\begin{aligned} f \frac{\Delta T_i}{\Delta r_i} &= \frac{\Delta T_o}{\Delta r_o} \\ \frac{\Delta r_i}{\Delta r_o} &= f \frac{1 - T_{\text{cell}}}{T_{\text{cell}}} \end{aligned}$$

For each of the two layers, we can prescribe a layer-specific *Rayleigh* number accordingly:

$$\begin{aligned} \text{Ra}_o &\propto T_{\text{cell}} \Delta r_o^3 \\ \text{Ra}_i &\propto (1 - T_{\text{cell}}) \Delta r_i^3 \end{aligned}$$

Having maintained non-dimensionality throughout, it is simple relate these two boundary *Rayleigh* numbers to the bulk Ra value:

$$\text{Ra}_{\text{layer}} = \text{Ra} \cdot \Delta T_{\text{layer}} \cdot \Delta r_{\text{layer}}^3 \quad (3.68)$$

At this point, however, we have exhausted the insight we can obtain without making further assumptions. If we provide that the inner and outer boundary thicknesses must be the same, as they are in the planar case, we can see that:

$$T_{\text{cell}} = \frac{f}{f + 1} \quad \leftarrow \Delta r_i = \Delta r_o \quad (3.69)$$

This, however, would imply that the inner and outer *Rayleigh* numbers are divergent. If we instead choose to conserve Ra , then: [‘Jarvis1993-cb]

$$T_{\text{cell}} = \frac{1}{1 + f^{-3/4}} \quad \leftarrow \text{Ra}_i = \text{Ra}_o \quad (3.70)$$

Both possibilities converge on 0.5 when $f \rightarrow 1$ and 0 when $f \rightarrow 0$, as we would expect.

However it is estimated, it is clear that, as Ra increases and boundaries thin, more of the mantle will fall in the intracellular region and global temperatures as a whole will approach T_{cell} . Conversely, if Ra slips below its critical value, the boundary layers will disappear and the entire domain will enter the conductive regime: $T^{av} = T_c$. These two temperatures therefore make up respectively the lower and upper endmembers of global temperature:

$$\begin{aligned} T_{\text{av}} &\approx T_c, & \text{Ra} < \text{Ra}_{\text{cr}} \\ &\rightarrow T_{\text{cell}}, & \text{Ra} \rightarrow \infty \end{aligned}$$

It makes intuitive sense that the effect of increasing Ra should be to decrease global temperatures, since that is exactly why convection is preferred wherever possible - though this intuition may not hold for all rheologies.

Of course, what we desire most of all is a cylindrical scaling for the mantle convection power law $Nu \propto Ra^{*\beta}$. Following [Jarvis, 1993] and mandating equality of inner and outer Ra_{layer} , it is possible to construct a ‘geometric correction’ $g(f)$ that functions as a coefficient of the *beta* scaling:

$$g(f) = \frac{Nu_c}{T_{\text{cell}}^{4/3}} \quad \leftarrow \text{Ra}_i = \text{Ra}_o \quad (3.71)$$

$$Nu = g(f) \cdot Ra^{*\frac{1}{3}} \quad (3.72)$$

Using this scaling, Jarvis was able to obtain a *beta* exponent of 0.321 ± 0.001 across four values of f from $(1.0 - 0.1)$ [Jarvis, 1993].

3.3.3 Internal heating in the annulus

It was established previously that, for a purely conductive internally heated system, the geotherm and geothermal gradient are represented by:

$$\begin{aligned} T_c(h) &= \frac{H}{2} (1 - h^2) \\ \frac{dT_c}{dh} &= H \cdot h \end{aligned}$$

This is intuitive because the source flux visible to each layer is proportional to the area below that layer, which goes linearly with height h in a planar domain.

In the annulus, though, the proportion of the domain beneath a given height h is instead represented by *Disc*, as we have shown. If we further assume that H is non-dimensionalised so as to represent the total flux of the model (i.e. it equals 1 for all geometries), then the flux through each layer height h of the annulus must simply be:

$$\phi_q(h) = H \cdot \text{Disc} \quad (3.73)$$

We show that this holds exactly Figure 3.8.

As before, the geothermal gradient required to transmit this flux must account for the varying layer length s^* - a function of h and the f parameter. Thus:

$$\begin{aligned} \frac{dT_c}{dh} &= -\frac{\phi_q}{s^*} = -H \frac{\text{Disc}}{s^*} \\ &= -\frac{H}{2} \frac{1+f}{1-f^2} \frac{(r^*+f)(r^*-f)}{r^*} = -\frac{H}{2} \frac{h(r^*-f)}{r^*} \\ &= -\frac{H}{2} \frac{h(fh - 2f - h)}{(fh - f - h)} \end{aligned}$$

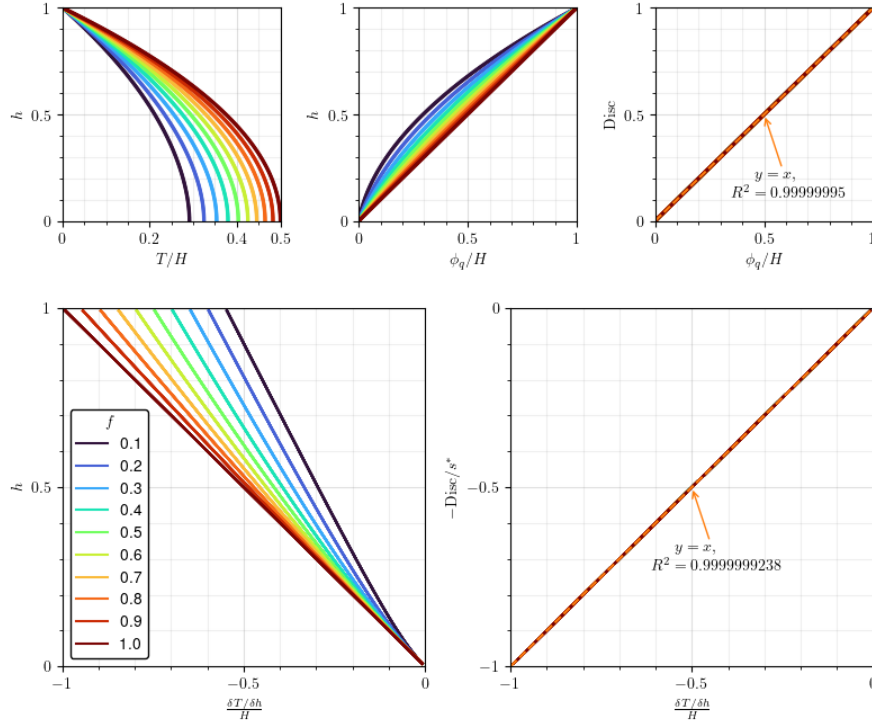


Figure 3.8: Summary of the scaling behaviours of isoviscous conduction under internal heating H for varying curvature parameter f (colours as in previous charts). While samples of varying heat have been plotted, they do not appear in these charts due to the intentional factoring out of H , demonstrating that this parameter is a simple coefficient.

To put it succinctly, the basic form of the relation is a rational function in terms of r^* :

$$\frac{dT_c}{dh} \propto \frac{r^* - f^2}{r^*} \quad (3.74)$$

The integral with respect to $h \in [0, 1]$ with $T(0) = 1$ yields the geotherm:

$$\begin{aligned} T(h) &= \frac{H}{4(f-1)^2} \left[2f^2 \ln |fh - f - h| - (fh - f - h)^2 + 1 \right] \\ &= -H \left(\frac{r_o}{2} \right)^2 \left(2f^2 \ln |r^*| - r^{*2} + 1 \right) \end{aligned}$$

Note how the correct choice of coordinate system dramatically simplifies things.

3.3.4 Mixed heating in the annulus

Like in the Cartesian case, the annular mixed heating regime contains both the internally-heated and basally-heated endmembers. The system reproduces basal heating in the trivial case of $H = 0$. The internal heating endmember arises in the dynamic case where the heating rate is at its ‘critical’ value, H_{cr} . At this exact value, the temperature in a layer infinitely close to the mantle base is equal to the fixed temperature of the mantle base proper. Consequently the flux across the boundary drops to zero, just as it would in the (basally insulated) internal heating case.

At values of H away from the critical value, there is always some non-zero flux across the lower boundary. The flux may be positive (heat flowing *into* the mantle) or negative (heat flowing *out* of the mantle). In either case we can say for sure that:

$$\phi_o + \phi_i + H = 0 \quad (3.75)$$

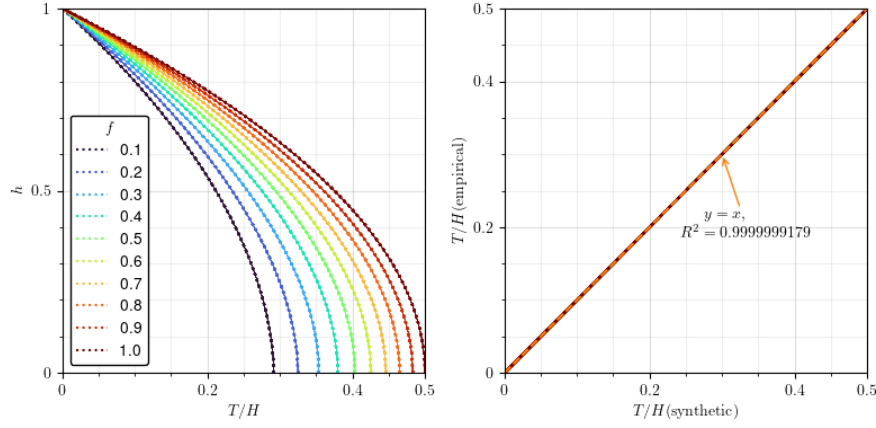


Figure 3.9: The conductive geotherm for cylindrical domains under internal heating. The empirical results and the symbolically-derived closed-form solution match exactly.

This is true whether H is high or low, or indeed, whether the mantle is conductive or convective. Let us consider the purely conductive case for now.

As before, we would like to obtain an exact closed-form solution for the conductive geotherm and average temperature. The value of H_{cr} is fully dynamic in the case of a mobile fluid, but in the purely conductive state, it will have a fixed value which is some function of the fixed model parameters. It would be good to obtain an expression for this too.

The first step, as previously, is to convert the empirical temperature profile into a geothermal gradient by taking a differential, then convert that gradient into the (axisymmetric) heat flux ϕ by multiplying by the non-dimensionalised height-dependent angular length s^* . Once we have the flux, we can take another differential to obtain the gradient of the flux.

$$\phi_{q_o} = H + \phi_{q_i} \quad (3.76)$$

In the Cartesian case, H_{cr} for mixed heating is equal to the inverse of the average temperature for purely basal heating. If the same holds for the annulus, we should therefore expect:

$$T_{\text{av}} = \frac{\sqrt[\varepsilon]{f}}{2}$$

$$H_{\text{cr}} = \frac{2}{\sqrt[\varepsilon]{f}}$$

The annular case also contains the Cartesian case as an endmember when f approaches 1. This gives us some clues about the limiting behaviour of the conductive geotherm for annular systems.

The existence of H_{cr} effectively splits the conductive regime into two separate subregimes. The low- H subregime is ‘monocooling’: only the outer boundary cools the system. The high- H subregime is ‘duocooling’: both boundaries cool the system.

Let us first consider the monocooling subregime. In these cases, all the flux through lower layers must pass through upper layers and ultimately through the upper boundary wall. The situation is comparable to the basally- and internally-heated cases, except with the addition of the lower boundary flux itself. If we can figure out what this is, it can simply be added to the flux for each subsequent layer all the way to the top of the domain. At $H = 0$, the lower boundary flux is trivially -1. As H increases, the total system temperature rises relative to the fixed lower boundary temperature and suppresses the flux across that boundary. At $H = H_{\text{cr}}$, the flux should come to zero by definition. The lower boundary flux is therefore constrained in this subregime to the range $[-1, 0]$.

We know that the flux through any given layer due to internally-generated heat alone should come to $\phi = -H \cdot \text{Disc}$ - i.e. each layer must transport the entirety of the heat produced by all lower layers. We can adjust this to account for the lower boundary flux ϕ_i , whatever it is:

$$\phi = -H \cdot + \phi_i \quad (3.77)$$

Basal heating in the Cartesian:

$$\begin{aligned} T(h)' &= -1 \\ T(h) &= 1 - h \end{aligned}$$

Internal heating in the Cartesian:

$$\begin{aligned} T(h)' &= -H \cdot h \\ T(h) &= \frac{H}{2} (1 - h^2) \end{aligned}$$

Mixed heating in the Cartesian:

$$\begin{aligned} T(h)' &= -H \left(h - \frac{1}{2} \right) - 1 \\ T(h) &= \frac{Hh}{2} (1 - h) - h - 1 \end{aligned}$$

Basal heating in the annulus:

$$\begin{aligned} T(h)' &= \frac{f - 1}{r^* \ln f} \\ T(h) &= \frac{\ln r^*}{\ln f} \end{aligned}$$

Internal heating in the annulus:

$$\begin{aligned} T(h)' &= -H \frac{\text{Disc}}{s^*} \\ T(h) &= -H \left(\frac{r_o}{2} \right)^2 \left(2f^2 \ln |r^*| - r^{*2} + 1 \right) \end{aligned}$$

3.4 Numerical methods and the Underworld code

When the font of analysis is exhausted, numerical solutions become indispensable. Though the construction of robust and accurate systems for the computation of mantle convection problems is far from trivial, many successful approaches have been developed over the years, each with its own advantages and limitations. For this thesis we adopt a finite-element approach, which has been extensively developed by many groups over more than a quarter of a century. In particular, we follow after Professor Moresi and colleagues, the developers of modelling codes including *CITCOM*, *Ellipsis*, and most lately *Underworld* [Moresi and Solomatov, 1995, Moresi et al., 1996, Zhong et al., 1998, Moresi et al., 2002, 2003, Farrington et al., 2005, Moresi et al., 2007, May and Moresi, 2008, Beucher et al., 2019].

Here we will discuss in broad terms the principles and infrastructure of our numerical modelling practice; particulars of model design and construction will be discussed in further chapters as relevance dictates.

3.4.1 Numerical methods

To take a numerical approach to mantle convection is to endeavour to iteratively solve the following system of equations, which are the Stokes, conservation, and advection-diffusion equations under the assumptions of incompressibility and infinite Prandtl number:

$$\begin{aligned}\nabla p - \nabla (\eta D) &= \Delta \rho \bar{g} \\ \nabla \cdot \bar{u} &= 0 \\ \frac{\partial T}{\partial t} + \bar{u} \cdot \nabla T &= \kappa \nabla^2 T + H\end{aligned}$$

Where η is dynamic viscosity, D the strain rate tensor, p dynamic pressure, $\Delta \rho$ the density anomaly, $\bar{g}$ the gravity vector, $\bar{u}$ the velocity vector, T temperature, κ thermal diffusivity, t time, and H a thermal source term, i.e. radiogenic heating.

The first two equations together solve for velocity given buoyancy and viscosity without accounting for inertia (hence the lack of time-dependency), while the third equation governs determines the change in temperature at any given point after an infinitesimal time interval due to diffusion (the first term) and advection (the second). Temperature, being the only time-dependent quantity, is the only necessary variable of state for this system. It is further possible to construct the viscosity and diffusivity parameters as functions of space, time, temperature, velocity, et cetera, to achieve much more complex rheologies as desired; plastic, elastic, rigid, and insulating materials may all be implemented in this way, each with its own perils and caveats.

We have already established that the equations are generally insoluble; numerical methods are the only recourse. There are unavoidable tradeoffs in terms of time, memory, accuracy, flexibility, extensibility, and robustness associated with every numerical convection scheme. Many methods have been developed, including smooth particle hydrodynamics [Monaghan, 2005] and discrete elements [Cundall and Strack, 1980]. We employ a finite elements approach [Hughes, 2012] in which the domain is divided into a discrete network of cells (the ‘mesh’): all field values are hosted on the nodes, with integrals across elements approximated using Gauss quadrature [Swarztrauber, 2003]. Such an approach can of course yield only an approximation of what the underlying equations imply, but the approximation can be arbitrarily accurate depending on the fineness of discretisation. The question, then, is how coarse can the approximation be made without losing fidelity to the governing equations?

Computationally, the problem takes the form:

$$\begin{aligned}A\bar{u} + B\bar{p} &= \bar{f} \\ B^T\bar{u} &= 0\end{aligned}$$

Where A is the known ‘stiffness’ matrix equivalent here to viscosity, $\bar{u}$ is a vector of unknown velocities, B is the discrete gradient operator, $\bar{p}$ contains the pressure unknowns, $\square^T$ indicates the transpose, and $\bar{f}$ is the known vector of body and boundary forces acting on the material. The objective is to solve for $\bar{u}$ as cheaply and reliably as possible; once a velocity solution is obtained, the system can be integrated in time and the cycle begins again. First, however, we require a solution for p . Multiplying both sides by $B^T A^{-1}$ and substituting for $B^T \bar{u} = 0$ we find:

$$B^T A^{-1} B \bar{p} = B^T A^{-1} \bar{f} \tag{3.78}$$

Such a form exposes the pressure to solution by a Schur Complement method, at the cost of introducing the non-trivial A^{-1} operation, which is both heavy and costly.

To avoid having to generate A^{-1} , we may instead reproduce its effect using a *Krylov Subspace* method (*KSP*), wherein matrix-matrix products are resolved by decomposing them into iterative series of matrix-vector products, with each vector comprising the residual of the previous iteration until the residual is less than some nominated threshold. One very popular implementation of this concept is the *Generalised Minimal Residual* method, or *GMRES* [Saad and Schultz, 1986]. Once a

solution for p is obtained, we are free to solve for u through similar methods; and this solution in turn may be used to advect the temperature field and any other state variables using a standard *Petrov-Galerkin* scheme; for convection problems an ‘upwind’ variant can be used to avoid ‘wiggles’ between nodes [Brooks and Hughes, 1982]. The actual time integral is carried out using a *Runge-Kutta* method for accuracy [Hairer et al., 2006] and is taken over a time interval chosen to be shorter than either the diffusive or the advective timescales across each element, i.e. the *Courant* condition [Courant et al., 1967]. Once advection has been carried out, the system is fully iterated in time and the process may be repeated, with the advantage that the pressure solution for the previous timestep can be retained to quicken the convergence of the subsequent step. Performance can be improved even further by incorporating preconditioners for each solve, which exploit *a priori* analytical insights into certain model configurations that may constrain the solution space [May and Moresi, 2008].

This approach is accurate, robust and stable [Moresi and Solomatov, 1995]. However, even in the best case scenario, the complexity of such a direct solution scales with the cube of the number of elements N^3 . To achieve a scaling behaviour closer to N , it is possible to drastically reduce the workload of the inner *KSPs* using an adaptive multi-grid approach. With this method, a solution is obtained first for a much coarser discretisation, and then corrected over successively finer meshes until the error is within a provided tolerance. In addition to speed and resilience, the multi-grid lends itself well to parallelisation, as the first-order global features of the model can be coordinated at the coarsest levels first, while finer local features captured in heavier arrays can be shared more judiciously, thus reducing communications overheads.

The method we have outlined is the product of decades of meticulous development and is known to be as reliable as it is quick. It has been comprehensively benchmarked across a wide range of rheologies and parameters, including models with extreme viscosity contrasts [Moresi et al., 1996], elastic behaviour [Moresi et al., 2002], and strain-localising mechanisms [Moresi et al., 2007]. It has been tested against physical laboratory experiments [Mériaux et al., 2018] and has been demonstrated to be robust and scalable up to thousands of parallel processes [Farrington et al., 2005].

One shortcoming of the method we have described is the inappropriateness of the finite element mesh for preserving the geometries of integer-valued domains - for example, deformation history, or the distribution of various material phases. Though there are ways of accommodating such features using mesh-based approaches, they tend to be cumbersome and inefficient. A superior approach is to incorporate Lagrangian particle swarms [Moresi et al., 2003], which are able to carry much higher-resolution information than the mesh and can transport both real-valued data (e.g. temperature, viscosity) and integer-valued data (e.g. history, material identity), which mesh-based variables necessarily cannot. Swarms and their associated variables are advected according to the Stokes solution at the same time as the underlying mesh-based state variables. In turn, the Stokes solver calls for the swarms to be interpolated to the mesh if and when they become relevant to the solution. Interpolation can be costly and complicated for unstructured networks like particle swarms - prohibitively so if the node weightings must be recalculated with each timestep, as is the case for any advecting swarm - so it is important to carry out such an operation as infrequently and efficiently as possible. Gauss quadrature is the standard method, with simple nearest-neighbour evaluation to determine which elements own which particles. An alternative is to use a grid-based Voronoi algorithm [Velić et al., 2009] in which domains of control for each node are iteratively built out cell by cell; where two nodes lay claim to the same territory, the interpolation grid is refined, but only inside the conflicted cell, thereby minimising superfluous calculations. Another strategy, particularly suited to the interpolation of very sparse swarms, uses a k - d tree to efficiently seek out the nearest particle from any given node. Regardless of interpolation style, the addition of particle swarms dramatically extends the utility of the finite element method.

3.4.2 The Underworld code

Particular software implementations of the methods outlined above have been developed over many computing generations, and several continue to co-exist today as part of a broad and branching family. The present state-of-the-art iteration is *Underworld*, which supports 2D, 3D, multigrid, and particle-in-cell features while also combining a powerful yet modular C-level infrastructure [Quenette et al., 2007].

with a user-friendly, hyper-extensible *Python*-based API. Parallelisation is provided through *MPI* while the underlying solvers are implemented with *PETSc*. Though deeper layers of Underworld remain fully transparent and accessible, the *Python* layer is designed to encourage fluidity, creativity, and legibility in model building, providing encapsulated higher-level proxies for the multifarious underlying *C* assets while subtly encouraging a good modelling idiom in users. This has encouraged bespoke application development [Beucher et al., 2019] and the integration of geodynamics codes with other modelling packages [Asten, 2018].

The higher-level Underworld syntax fully exploits *Python*’s object-oriented character, encapsulating standard model features like meshes, swarms, variables, and solvers as independent instances of generalised classes. Each object corresponds to *C*-level structures which are in turn organised under the *StGermain* interoperability framework for computational modelling [Quenette et al., 2007]. The algorithmic firepower at the heart of the operation draws on the ubiquitous standard *PETSc* code for partial differential equations. By default, the *PETSc* infrastructure is configured for robustness first, speed second; however, a range of options is exposed at the *Python* level to reconfigure the solvers as desired. The principles of encapsulation and localisation are carefully honoured in Underworld’s design, with use of global attributes minimised and namespace pollution strictly avoided. Consequently, deletion of obsolete references or ‘garbage collection’ operates mostly as a *Python* user would intuitively expect, so that in typical use cases it is rarely necessary to do more than the elementary due diligence to limit memory leaks. However, at the scales we have operated at, unavoidable pointer entropy at the *C*-level has been found to proliferate to problematic levels at times. This has been mitigated by prudent reuse of already instantiated objects and by spawning the heaviest or lengthiest jobs in subprocesses to harness system-level garbage collection.

As accessible as the new tools are, care must still be taken to ensure an appropriately configured model, beginning with the choice of resolution. While temporal resolution, i.e. timestep size, is determined dynamically based on the prescribed tolerances, spatial resolution is the domain of the user. Overly fine elements are wasteful, while insufficiently fine elements will lossily discretise the underlying physics. As a rule of thumb, the spatial resolution in any given region should be half an order of magnitude finer at least than the smallest relevant model features in that vicinity. Of course, it is not always clear *a priori* what this scale will be. Boundary layer and plume theory can provide some information about featural dimensions at steady-state for simple rheologies; however, the sensitivity of mantle convection on potentially very small-scale instabilities means a resolution sufficient for a steady-state solution may still bias the solution at other stages. The appropriate resolution can be sought empirically, by running a suite of progressively finer models until the point of diminishing returns is reached. In theory, a well-constructed model should converge with resolution in the limit that a discretised model becomes indistinguishable from a continuous one. If convergence does not occur within a computationally feasible envelope, the model may be presumed to be misconfigured in some deeper sense. Another means of determining the correct resolution is to run a single, very-high resolution test and conduct a power spectral analysis of the constituent fields; the shortest wavelength that contains information must dictate the spatial resolution. The nexus of *Rayleigh* number, featural thickness, and resolution places an upper bound on what parameters can realistically be tested in the context of a large suite-modelling experiment: although *Ra* values approaching 10^9 are likely more appropriate for Earth whole-mantle convection [Wolstencroft et al., 2009], most cases explored here are less than $Ra = 10^7$, which is amenable to resolutions in the order of 128 radial cells. If resources are particularly at a premium, static or even adaptive mesh refinement can be employed to concentrate resolution in areas where it is most needed. Unfortunately, it is a present shortcoming of Underworld that only quadrangular elements are supported, starkly limiting the options for grid refinement. Work toward supporting unstructured meshes is, however, underway.

Underworld provides an easy interface to the underlying *PETSc* options, most notably the choice of inner solve method and tolerance. The default configuration for the solver is *mg* or ‘multigrid’, which is the method we have outlined here: this arguably provides the best balance of speed, robustness, scalability, flexibility, and parallelisability. Alternatives include *mumps*, ‘multifrontal massively parallel sparse direct solver’, and the ‘lower-upper method’ LU. Careful benchmarking is called for to choose the correct configuration, and optimum results cannot be assumed for the default configuration. Toler-

ances in particular should always be calibrated manually using convergence and power spectrum tests as outlined above. Excessively fine tolerances will needlessly delay solver convergence, while overly generous tolerances will introduce numerical noise into the solution. The chosen tolerances ultimately determine the uncertainty inherent to the model and should always be chosen with care.

3.4.3 Underworld in the annulus

Even with the gift of Moore’s law, fully three-dimensional models remain prohibitively expensive, particularly at sufficient resolutions. A two-dimensional annulus is a compromise that allows exploration of the influence of curvature without exponentially expanding the degrees of freedom. Although such a geometry cannot claim to reproduce Earth-like conditions as such, it is nonetheless appropriate for probing a wide range of mantle convection phenomena and continues to be widely used.

In Underworld, which at present uses Cartesian-type meshes, the annulus is constructed by deforming a rectilinear mesh around the origin such that the ratio of inner and outer radii f falls in the range $0 \rightarrow 1$, where $f \rightarrow 1$ approaches no curvature and $f = 0$ is a model with no core. When the aspect ratio and curvature are such that the two ends meet, those ends are made periodic and the result is a full annulus. For the whole Earth mantle, a ratio of $f = 0.54$ is appropriate; for the upper mantle only, a ratio of $f = 0.9$ is more realistic. Naturally, a full annulus of $f = 1$ is not possible, while severe solver complications manifest at $f < 0.2$ due to extreme elongation of the basal cells.

Our approach to the annulus has the advantage of robustness and simplicity. It is more amenable for the solver and allows highly curved geometries without requiring a revision of any of the critical computational systems. However, it does have many shortcomings. Outer cells are stretched as inner cells are shortened, forcing a choice of either under- or over-resolution of one or other of the boundary layers. Boundary layer conditions for velocity must be defined according to unit vectors rather than simply Cartesian components; it then becomes necessary to rotate and unrotate the boundary vectors during each solver loop. In periodic cases with zero-shear upper and lower boundaries, we must also be careful to suppress any solid-body rotation that might emerge by calculating and subtracting any uniform global angular components from the velocity vector field. All of this takes time and CPU cycles, not to mention an increasingly sprawling code overhead.

For our annulus models, it has proven useful to write a bespoke mapping protocol to go from Cartesian to annular domains. The ‘box’ algorithm projects a standard unit square of $x, y : (0, 1)$ onto each annulus or annular wedge such that Cartesian positions can be smoothly and swiftly evaluated in the box and vice versa. This provides a common frame of reference between rectilinear and curvilinear models of any scale and dimensions, with benefits for interoperability, usability, and visualisation. The procedure is both parallel-safe and parallel-efficient, and approaches *C*-level performance through careful, idiomatic usage of the NumPy interface. In theory it could be extended to any geometry which is a continuous mapping of a rectilinear mesh.

3.5 Conclusion

In this chapter we have discussed the theory of mantle convection; explained and justified our preferred numerical approach; introduced the software tools requisite for the job; and presented a detailed conceptual framework for our research methodology. Further and more detailed methods will be explained in future chapters as they become relevant. Although the challenge at hand is formidable - ever more evidently so as we progress - we hope we have demonstrated that the methodologies thus far outlined are tried, true, and fit for purpose. With equal parts ambition and caution, let us proceed.

Chapter 4

Methods

#####

In Western military theory, there is a maxim: “Quantity has a quality all its own.”

In the natural sciences, this maxim has a double-edged truth. Quantity - of observations, of experiments, of models - opens up new frontiers of discovery: not greater per se, but profoundly different to the frontiers of ‘small-run’ science. At the same time, quantity creates new technical challenges: again, not necessarily greater, but fundamentally different to the challenges of executing a narrowly-focussed study.

To the old maxim, we might add a corollary: “And different quantities may have different qualities too.” In other words, not all forms of “big” science are created equally. The 2010s saw a boom in ‘Big Data’. The promises made at that time were as sweeping and ecstatic as those attending today’s AI boom. In many cases, however, the promised transformations were abortive, underwhelming, or simply not forthcoming.

Moving between scales successfully requires understanding what properties are preserved, lost, and changed in the scaling. It requires tools that are responsive to the unique qualities of working at a particular scale, and an appropriate mentality on the part of the scientist to recognise the costs and pursue the novel benefits on offer.

Geodynamics has always been at the vanguard of new methodologies, embracing numerical modelling, Big Data, and now AI, with energy and felicity. This thesis aims to contribute to that legacy by exploring the potential of highly object-oriented approaches to open up new ways to ‘go big’ on the model frontier.

At its simplest, this thesis is committed to exploring the potential benefits of a simple tradeoff between two kinds of ‘quantity’: lower resolution (quantity of calculations per model) in exchange for richer parameter space coverage (quantity of cases sampled). It will be shown how the journey from small-batch modelling to big-batch and, finally, truly massive suite-modelling, has entailed radically new approaches to designing, managing, shepherding, storing, organising, analysing, visualising, and even conceptualising big model runs for the big computing age - and not just for geodynamics, but for any field targeting similar ‘shaped’ problems.

In this chapter, we will first briefly sketch the problem at hand, then develop a formalisation of our approach to the problem; next, we will present our two research softwares: Everest, a general-purpose research data framework, and PlanetEngine, an Everest ‘engine’ built for our particular subject matter of planetary geodynamics; finally, we will show the code in action and discuss the future trajectory of the program.

#####

Scraps

The downside of being in the vanguard - to continue the military metaphor - is that one is invariably among the first to crash into unforeseen obstacles: often painfully. The wider world has been the beneficiary of planetary scientists’ willingness to commit substantial resources to deep methodological innovation: particle-in-cell methods, inverse modelling, and Krige-type interpolation are all owed to the field.

#####

4.1 The crisis of dimensionality

How do you model an infinite frontier?

Most scientific modelling is focused on deeply exploring a tightly-chosen set of scenarios that relate to a real or hypothetical situation in some specific way. For that reason, it has traditionally sufficed to treat models as a kind of laboratory apparatus, even when the model itself is a numerical (software) entity that is not constrained in the ways that physical experiments are.

In this way of thinking, a model is a kind of machine, no different from a mass spectrometer or a centrifuge. Empirical data goes in one end, the machine is set to ‘run’, and model outputs pop out the other end. The analogy with conventional industrial processes is neither inapt nor accidental: the scientist features in this story as just another kind of technician on another kind of assembly line; the model is capital.

Just as industrial innovations have unlocked new scales of production - and thereby, new classes of product - so can innovations in the field of modelling expose new avenues of discovery. As in manufacturing, the frontiers stretch away from us on three (not entirely orthogonal) axes:

- Space: meaning actual physical space, but more acutely, factors of production that are embedded in space, like computational hardware, network infrastructure, and real estate.
- Time: meaning actual physical time, but more acutely, factors of production that are embedded in time, like CPU cycles, read-write time, labour time, and all sorts of *pro rata* costs, like electricity, water, depreciation, wages, and so on.
- Mass: the general ‘systems complexity’ of the model considered as a whole, including its human user; difficult to quantify, but related to codebase size, Big O analysis, Kolmogorov complexity, and Shannon entropy.

In the early twentieth century, a wave of ‘Time and Motion’ studies saw researchers dispatched to factories to document exactly how each action of each worker contributed to the final product. Let’s set ourselves as a fly-on-the-wall of a typical worker in a modern ‘science factory’. How does the labour of a modern modeller change as the task evolves across our three ‘axes of complexity’ - and where might better tooling make a decisive difference?

4.1.1 Mass

One-shot modelling The simplest kind of modelling you could do is a ‘one-shot’ model. Performing a regression analysis on a collected dataset is one example: the inputs here are the data to be analysed; the outputs are the parameters of the fit, including the error.

The ‘workflow’ is something like this:

One-shot scientific modelling workflow

1. Marshal the input data.
2. Configure the apparatus.
3. Feed in the input data and hit ‘run’.
4. Wait.
5. Collect the outputs.

Given that the model apparatus in principle acts on generic data, there is nothing stopping us from feeding the outputs into another model, and another, forming a ‘model pipeline’: the data equivalent of a physical assembly line. An unskilled operator might carry this out manually, paying a labour cost at each hand-off; a more skilled operator will recognise the redundancy and ‘compose’ these models into a single compound model, within which the component models are executed sequentially. In either case, the process remains, conceptually, a ‘one-shot’.

Preparing a one-shot model is easy. The data must be made available in the right environment and in the right format. Crucially, it can be stored under a completely arbitrary name - as can the output, and for that matter, the model itself. (We will later see how big an advantage this is.) If we want to guarantee reproducibility, we only need to tag the output with the input name and the model name. If we want to publish our work, we merely prepend our own name to the whole assemblage to make it universally disambiguous.

In terms of computer resources too, a one-shot is as convenient as can be. The input footprint is known. We may not exactly know what the runtime and output footprints will be, but we can usually guess. The amount of (machine) time required to run the model may not be known, but we can set up an alert to inform us when it is done, and in the meantime deploy our own labour gainfully in some other place. The labour-time running cost is therefore close to zero.

Without going out and doing a huge anthropological survey of all science, we cannot put specific numbers on how common this sort of workflow is: certainly, the number will vary between disciplines. For the field of geodynamics, we can say that it is very common indeed. We can see that a one-shot study has several virtues: it is easy to organise, easy to run, easy to comprehend, and easy to explain. The model itself may be complicated and costly - but up to a certain point, these are costs born by machines, not by humans. In a market where skilled labour is expensive and infrastructure is cheap (or, almost equivalently, in which labour is risky and infrastructure is safe), economic forces will militate in favour of the one-shot.

4.1.2 Space

What if the input data we are dealing with is not megabytes, or gigabytes, but rather terabytes - or even petabytes? What if the process demands not a handful of CPUs, but a warehouse-full?

The ‘space’ axis dictates what sort of machine we need to actually execute our model:

- Laptop science: fits comfortably on a good-quality laptop; a dozen gigabytes of RAM, less than a terabyte of storage, a dozen CPUs and/or a small GPU; thousands of dollars of hardware.
- Workstation science: fits on a powerful desktop machine; up to a hundred or so gigabytes of RAM, many terabytes of storage, dozens of CPUs and/or several top-shelf GPUs; tens of thousands of dollars of hardware.
- Node science: too big for any one physical machine, but still small enough that it can run on a single virtual machine, emulating a conventional operating system, but spread over multiple physical devices in a server cabinet; hundreds of gigabytes of RAM, dozens to hundreds of terabytes of storage, hundreds of CPUs and/or dozens of GPUs; hundreds of thousands of dollars of hardware.
- Server science: too big for a single stable VM, but small enough that it can run across a single tightly-connected physical network under a specialised operating system; petabytes of storage, terabytes or more of RAM, potentially thousands or hundreds of thousands of CPUs and GPUs; millions of dollars of hardware.
- Supercomputer science: too big for conventional computing altogether; exascale complexity footprints, hundreds of millions or billions of dollars of capital.

Note how each threshold introduces new infrastructure, new skills, new dependencies, and ultimately, new stakeholders to the project. Anyone can run any laptop model on any laptop, but it takes a skilled hobbyist to maintain a good-quality workstation and clever model design to exploit it. A node can be served by a generic VM provider *pro rata* without needing to involve anyone else in the project, but running a job on a whole server requires on-the-ground staff who are at least broadly aware of the needs and complexities of the model, not to mention highly specialised model architectures and the concomitant skills to drive them. Finally, supercomputer science demands tight coordination between user and provider and, typically, software that is specialised not just for supercomputers generally but for each supercomputer specifically.

4.1.3 Time

The ‘time’ axis introduces its own escalating ladder of difficulties:

- Trivial science: runs in the blink of an eye; e.g. drawing a scatter plot or averaging a spreadsheet.
- Hourglass science: takes long enough to run that the user becomes aware of it, but not so long that the user can background the process - e.g. loading a multi-gigabyte CSV file or rendering a 3D plot.
- Teabreak science: time-consuming enough that the user can safely background the process and gainfully focus on something else for a short time - e.g. running a clustering algorithm on a large dataset (while making a cup of tea).
- Redeye science: time-consuming enough that the user can safely change focus to an altogether different activity - e.g. running a mid-resolution fluid model (while flying overnight to a different country).
- Longhaul science: may take anywhere from a day to a week - e.g. running a regional climate model while preparing a paper on the topic.
- Slowburn science: could take weeks, months, or even years - e.g. running a high-resolution galactic evolution model while writing up a paper on a totally different topic.

The escalating cost of compute with time is obviously one dimension of time complexity, but the labour cost also shifts in complex ways. At the ‘trivial’ level, there is no labour cost at all. At the ‘hourglass’ level, the user’s time is mostly consumed (or rather, wasted) in tandem with compute time. From the ‘tearoom’ stage up, the user can make up time by switching tasks (though that incurs its own costs). At even longer time horizons, the assumption that human labour and machine labour run ‘in parallel’ starts to unravel. Long-running processes need shepherding: a mandatory update may trigger, or the power may go out, or some unforeseen resource limit may be stretched. The cost of mistakes and changes-of-mind also grows with time - potentially exponentially, if there are onward task dependencies; likewise the cost of bottlenecks, undershoots, and overshoots. At the higher end, the relationship of compute time and labour time inverts, as long-running projects demand increasingly vigilant stewardship.

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Scraps

In the early twentieth century, a wave of ‘Time and Motion’ studies saw observers dispatched to factories to document exactly how each individual worker contributed to the whole. Let’s set ourselves as a fly-on-the-wall of a typical scientific modeller.

(This economic force is demonstrated at scale in the modern AI sector, where each of the gigantic ‘foundation models’ (like GPT or Llama) is essentially the monolithic output of a single, multi-billion-dollar execution of a compound one-shot (the training pipeline) on a particular input set (the training data). That the output in this case is itself a model is an interesting detail we will pick up on later. The gargantuan size of these models creates ‘a quality all its own’ which .)

Chapter 5

Critical

5.1 Conclusion

Chapter 6

Temperature-dependant rheology

Chapter 7

Viscoplastic rheology

Chapter 8

New techniques

Chapter 9

Discussion

Chapter 10

Conclusion

A conclusion.

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